

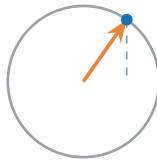
THE CURIOUS FAMILY LIBRARY

Mathematics · The Foundation

Numbers and Functions

MODULE ONE

*Where the numbers come from, what a function really is,
and why a single new number ties it all together.*



The first step of a mathematics foundation built to carry you into AI, engineering, medicine, science — or wherever your curiosity leads.

Companion notebooks (free in Google Colab) and extra materials:

github.com/MeiraDavidson/math_module1

For the Jessica family kids —

be curious, stay open, be kind.

The Mathematics Foundation

The *Curious Family Library* is a family of plain-spoken, build-it-yourself guides across many subjects. This book opens its **Mathematics** line — **Module One** of a small, deliberately ordered foundation. Its aim is easy to say and ambitious to do: to give you the **backbone of mathematics** — quickly, and deeply enough that it lasts — so you can dive into whatever you choose: artificial intelligence, engineering, medicine, science, or a field that doesn't even have a name yet.

The goal is never to memorize. It is to **understand well enough to rebuild**. Every idea here is earned from a reason, so you walk away with not a list of facts but two durable habits: the **critical thinking** to ask whether a claim is really true, and the power to **derive what you need** when you've forgotten the formula — which, in real life, is always. Master the foundation and you needn't carry the whole library in your head; you can *regenerate* it.

The modules of the foundation:

1. **Numbers and Functions** (*this book*) — the number systems, and the handful of function families that everything else is built from.
2. **Calculus I** — single variable: limits, derivatives, integrals — the mathematics of change and accumulation.
3. **Linear Algebra** — vectors, matrices, and the language of high dimensions. It sits here on purpose: Calculus II leans on it (and it is the language modern AI lives in).
4. **Calculus II** — several variables: the ideas of Calculus I set loose in more than one dimension, standing on the vectors and matrices just learned.
5. **Probability and Statistics** — reasoning carefully about uncertainty, data, and evidence.
6. **The Summary** — a short capstone that ties the whole foundation together and shows how the pieces connect.

They're meant to be read roughly in order — each leans on the ones before — but every module stands on its own feet. *Computing and algorithms* have a line of their own in the Library; this is the backbone they rest on.

A note before you begin

This is not a book to race through. It is a book to *argue with*.

Most maths books hand you rules and ask you to trust them. This one does the opposite: almost nothing here is something you have to take on faith. Every new kind of number, every formula, every strange-looking symbol arrives with a reason — usually because something simpler *broke*, and we had to invent a repair. If you keep asking “why did we need that?”, you will almost always find the answer a sentence or two later. That habit — expecting the pieces to connect — is the real subject of the book.

A few signposts you will meet along the way:

- ◆ **Why-it’s-true boxes.** The shaded boxes hold the deeper proofs. On a first read you may skip every one of them and still follow the whole story. Come back when you are ready to see *why*, not just *what* — that is where the real brain-training lives.
- ▶ **Run-it-yourself callouts.** A handful of ideas in this book are best *watched moving*. Where you see this mark, there is a short live notebook you can open and play with — no setup, just click and run.
- ★ **Teach-it problems.** Each problem set ends—no, *begins*—with a request to explain something out loud to a brother, sister, or parent. This is not filler. The moment you have to say an idea to another person, the gaps in your own understanding light up like warning lamps. It is the single most valuable thing you can do.

Companion materials. Several chapters have a short live Python notebook — the ▶ callouts above point to them — that you can open and run free in **Google Colab**. All you need is a Google (Gmail) account; there’s nothing to install and no other sign-in. Those notebooks, plus additional problem sets and worked solutions as they’re posted, live at github.com/MeiraDavidson/math_module1.

Take your time. Draw the pictures yourself. Explain them to someone. Welcome aboard.

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CHAPTER 1

Counting Isn't Enough

Here is something nobody tells you when you're little: the numbers you learned first were never the whole story. They were just the beginning — the part we needed before we needed anything else.

Let's start where everyone starts.

1.1 The numbers you already trust

You can count. One sheep, two sheep, three sheep. These are the **counting numbers**:

1, 2, 3, 4, 5, ...

They go on forever, and there's something satisfying about them. They feel solid. You can hold up that many fingers. You can put that many cookies on a plate. Mathematicians call them the **natural numbers**, because there's nothing artificial about them — they're the numbers that show up the moment you have *stuff* and you want to know *how much stuff*.

For a long time in human history, this was basically all the numbers there were.

But here's the thing about numbers: we don't just count with them. We *do* things with them. We add. We subtract. We multiply. We divide. And the moment you start doing things with numbers, you discover that your nice solid collection of counting numbers has holes in it.

Let me show you what I mean. This is going to be the secret pattern of the whole chapter, so watch for it.

1.2 When subtraction breaks

Adding two counting numbers is no problem. $3 + 5 = 8$. Still a counting number. You can add counting numbers all day and you'll never leave the family of counting

numbers. Nice.

Now subtract.

$$5 - 3 = 2.$$

Fine. Still a counting number.

$$3 - 5 = ?$$

Uh oh.

If all you have is counting numbers, this question has *no answer*. You're being asked to take 5 cookies away from a plate that only has 3 cookies on it. With actual cookies, you'd just say "you can't do that." And for thousands of years, that's exactly what people said. The problem was considered nonsense.

But think about how strange that is. We have a perfectly good operation — subtraction — and it just... refuses to work sometimes. It works for $5 - 3$ but not for $3 - 5$. That's annoying. An operation that only sometimes works is like a light switch that only sometimes turns on the light.

So mathematicians did something bold. Instead of saying "you can't do that," they said:

What if we invented the numbers that would make it work?

And that's where the negative numbers come from. We *decided* that $3 - 5$ should have an answer, and we gave that answer a name: -2 . We extended our collection. Now we have:

$$\dots, -3, -2, -1, 0, 1, 2, 3, \dots$$

This bigger family is called the **integers**. (Notice 0 snuck in there too — the number for "none," which is also a surprisingly modern invention, but that's a story for another day.)

Here's the part I really want you to feel. We didn't *discover* negative numbers lying around in nature. Nobody ever held up negative-two fingers. We **built** them, on purpose, because we wanted subtraction to always work. The integers exist because we refused to accept an operation that broke.

That's the pattern. Hold onto it.

1.3 When division breaks

Now let's do the same thing again, one level up.

Multiplication of integers? No problem. $4 \times 3 = 12$. Multiply integers all you like, you stay among the integers.

Division?

$$12 \div 3 = 4.$$

Fine.

$$12 \div 5 = ?$$

Hmm. If all you have is integers, this question has no answer. You can't split 12 into 5 equal whole pieces. With actual cookies, you'd be stuck — unless you're willing to break the cookies.

And of course you are. Of course you'd break the cookies. Each person gets two cookies and then a little bit more — two and two-fifths, in fact.

So once again, we have an operation — division — that refuses to always work if we only allow integers. And once again, instead of accepting a broken operation, we invent the numbers that fix it.

These are the **fractions**: numbers like $\frac{12}{5}$, $\frac{1}{2}$, $\frac{7}{3}$, $-\frac{4}{9}$. A fraction is just one integer divided by another (with the bottom one not zero — we'll get to why later). Mathematicians call these the **rational numbers**, and the word *rational* has nothing to do with being sensible. It comes from *ratio* — a fraction is a ratio of two integers, one number “on top of” another.

The rational numbers are a huge upgrade. Now both subtraction *and* division always work. Almost any arithmetic you can think of stays safely inside the rationals.

Do you see the pattern now? Let me say it plainly, because it's the most important idea in this chapter.

Every time we extended our numbers, we did it for the same reason: **an operation broke, and we built the numbers that would let it work again.**

- We added **negatives** so that *subtraction* always works.
- We added **fractions** so that *division* always works.

Each new family of numbers contains all the old ones and then some. The counting numbers live inside the integers, which live inside the rationals. We never threw anything away — we only ever added what was missing.

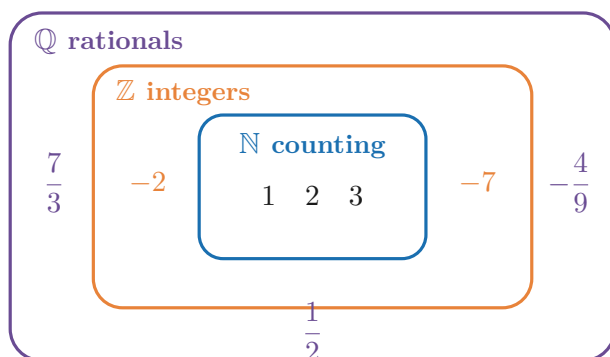


Figure 1.1. The nested number systems: $\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q}$. Each larger system contains all the smaller ones — we never threw anything away, we only ever added what was missing.

1.4 A picture: the number line

There's a beautiful way to see all of this at once. Draw a straight line. Pick a spot, call it 0. Pick another spot to its right, call it 1. Now you've got a ruler for the whole universe of numbers.

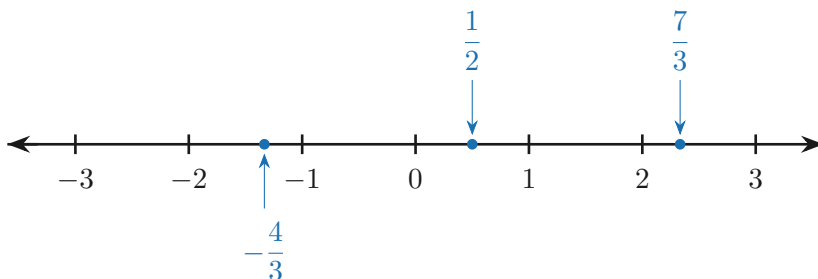


Figure 1.2. The integers are evenly spaced fence posts; the fractions live *between* them. Here $\frac{1}{2}$ sits exactly halfway between 0 and 1, $\frac{7}{3}$ a little past 2, and $-\frac{4}{3}$ between -2 and -1 .

The integers are evenly spaced fence posts marching off in both directions. The fractions live *between* the fence posts: $\frac{1}{2}$ sits exactly halfway between 0 and 1, and $\frac{7}{3}$ sits a bit past 2.

And here's something that feels almost magical about the rationals: between *any* two of them, no matter how close, there's always another one. Pick $\frac{1}{2}$ and $\frac{1}{3}$. Right between them sits their average, $\frac{5}{12}$. And between *those* two there's another. And another. Forever. The rationals are packed in infinitely tightly. There are no gaps — or so it seems.

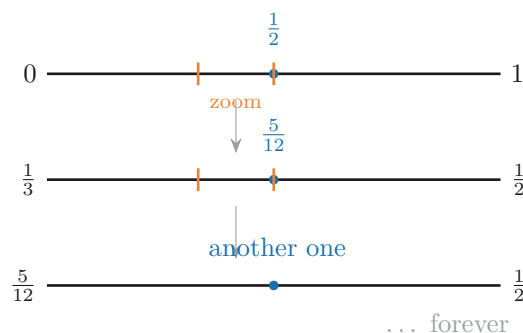


Figure 1.3. Density of the rationals. Between any two fractions there is always another — and between those, another still. Zoom in as far as you like; the packing never runs out.

Sit with that for a second. Between any two fractions, however close together, you can always squeeze another fraction. It feels like the rationals must fill up the entire line, leaving no room for anything else.

That feeling is wrong.

And the fact that it's wrong — the discovery that the number line has holes in it even after you've packed in every possible fraction — is one of the most shocking things in all of mathematics. It broke the people who first found it. There's a legend that the man who discovered it was thrown off a boat for telling the secret.

That's the next chapter.

Things worth thinking about

1. We invented negatives to make subtraction always work, and fractions to make division always work. Addition and multiplication never needed fixing. Why do you think *those* two operations were already fine, but their “opposites” caused trouble?
2. We said division always works for rationals — *almost*. The one thing you still can't do is divide by zero. Try it: what number could $\frac{1}{0}$ possibly be? See if you can talk yourself into an answer, and then see why it falls apart.
3. Between any two fractions there's always another fraction. Does that mean there are “more” fractions than integers? It seems obvious that there must be. Hold onto your answer — it turns out to be one of the great surprises of mathematics, and it's not what you'd guess.
4. Every time an operation broke, we invented new numbers to fix it. Can you

imagine an operation that even the *fractions* can't handle — something that would force us to invent yet another kind of number? (You already know one, secretly. Think about what number, times itself, gives 2.)

Problems to work

★ Teach it — explain it to someone

Problem 1 — Teach it (the invention pattern). Find someone at home — a brother or sister, a parent — and tell them the cookie story. Start with the plate that has 3 cookies and someone asking you to take 5 away. Walk them all the way to “so we *invented* a number, -2 , to be the answer.” Then do it again with sharing 12 cookies among 5 people, arriving at fractions.

The thing you are really teaching is the *pattern*: an operation broke, so we built the number that fixes it. If your listener can repeat that one sentence back to you, you've got it — and you'll never again think of negative numbers or fractions as things that were just “always there.” You'll know they were *built*.

Problem 2 — Sort the zoo.

Put each number into the *smallest* family it belongs to: counting number ($\mathbb{N}$), integer ($\mathbb{Z}$), or rational ($\mathbb{Q}$). Remember the families are nested, so “smallest” means the innermost ring it fits in.

$$7 \quad -4 \quad \frac{3}{8} \quad 0 \quad -\frac{10}{2} \quad \frac{9}{3}$$

(Two of these are sneakier than they look — simplify before you decide.)

Problem 3 — Squeeze one in.

For each pair of fractions, find a fraction that lies strictly between them. (The average — add them and halve — always works, but see if you can sometimes spot an easier one.)

$$(a) \frac{1}{4} \text{ and } \frac{1}{2} \quad (b) \frac{2}{5} \text{ and } \frac{3}{5} \quad (c) \frac{3}{7} \text{ and } \frac{4}{7}$$

Problem 4 — The divide-by-zero trap.

Suppose $\frac{1}{0}$ were some number — call it n . Then, just like $\frac{1}{2} = n$ would mean $1 = 2 \times n$, having $\frac{1}{0} = n$ would mean $1 = 0 \times n$. What does $0 \times n$ equal, for *any* number n ? Use that to explain, in one sentence, why no value of n can possibly

work — and why dividing by zero is banned, not by decree, but because it's genuinely impossible.

Problem 5 — [challenge] Always room for one more.

You can keep squeezing fractions between fractions forever, and Problem 3 hinted at why. Prove it in general: take *any* two different fractions $\frac{a}{b}$ and $\frac{c}{d}$, and show that their average $\frac{1}{2}(\frac{a}{b} + \frac{c}{d})$ is (i) again a fraction, and (ii) strictly between the two you started with. Once you've shown the average is always a brand-new fraction in the gap, explain why that means the squeezing can *never* stop. (This is your first real proof of something true for infinitely many cases at once — a taste of what's coming.)

Problem 6 — [challenge] Is there a smallest positive fraction?

A friend claims that $\frac{1}{1,000,000,000}$ is “basically the smallest fraction there is.” Show they're wrong by naming a positive fraction that is even smaller — and then explain why their search is hopeless: why there is *no* smallest positive fraction at all. (Hint: whatever tiny fraction someone hands you, what could you do to its bottom number?)

Problem 7 — [challenge] The next wall.

Every family we built repaired a broken operation. Here is a brand-new operation: “take the square root.” Experiment with a calculator: $\sqrt{9} = 3$ (a tidy rational), $\sqrt{\frac{1}{4}} = \frac{1}{2}$ (also tidy). Now try $\sqrt{2}$. Write down the first eight digits it shows you, then square that number by hand and see whether you land exactly on 2. You won't — and chasing *why* you can't is precisely where Chapter 2 begins.

CHAPTER 2

The Hole in the Number Line

At the end of the last chapter I made you a promise. I said the rational numbers — all those fractions packed infinitely tightly along the line — don't actually fill it up. I said there are holes.

Let me show you one.

2.1 A perfectly innocent question

Draw a square. Make each side exactly 1 unit long. Now draw the diagonal — the line from one corner to the opposite corner.

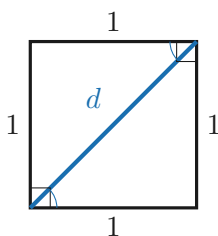


Figure 2.1. The unit square and its diagonal d . The diagonal cuts the square into two right triangles, each with short sides 1 — the perfect setup for Pythagoras: $1^2 + 1^2 = d^2$. The diagonal is a real, drawable length, so it *must* be a number.

How long is that diagonal?

This isn't a trick question. It's about the most ordinary shape there is. A square with sides of length 1. Surely its diagonal has *some* length. And surely that length is a number — a number we can write down as a fraction, like everything else we've met so far.

There's a famous fact — you may have already bumped into it — that tells us exactly how long the diagonal is. It's called the **Pythagorean theorem**, and for a right triangle it says: the two short sides, squared and added together, equal

the long side squared. Our diagonal cuts the square into two right triangles, each with short sides of length 1. So if we call the diagonal's length d :

$$1^2 + 1^2 = d^2 \quad \implies \quad 2 = d^2.$$

So d is the number whose square is 2. We write it $\sqrt{2}$, the **square root of 2**.

Now here's the question that's going to crack the world open. We *know* this length exists — you just drew it. So it must be a number. And every number we've met so far is a fraction. So:

What fraction is $\sqrt{2}$?

Take a moment and try to find it. Hunt for two whole numbers, one over the other, that give you 2 when you square them.

You might try $\frac{3}{2} = 1.5$. Square it: 2.25. Too big. Try $\frac{7}{5} = 1.4$. Square it: 1.96. Too small, but close! Try $\frac{17}{12} \approx 1.4167$. Square it: about 2.0069. So close!

Fractions creep toward $\sqrt{2}$ but never land on it

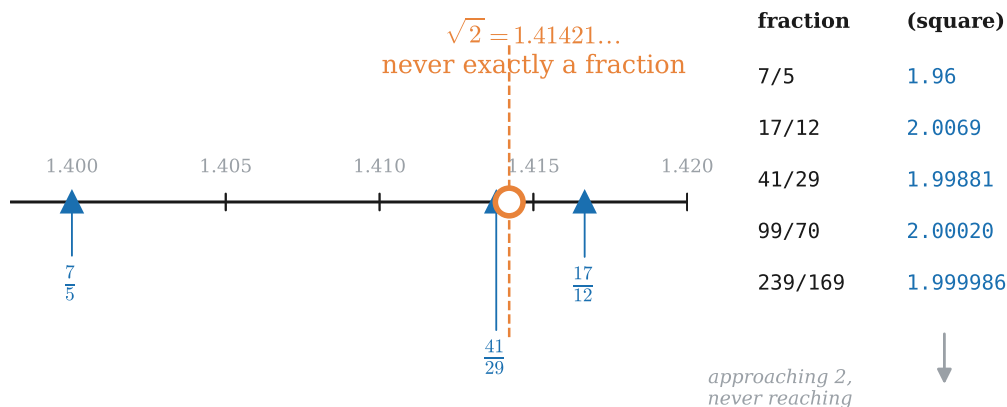


Figure 2.2. The hunt for $\sqrt{2}$ as a fraction. The convergents creep closer and closer from both sides, and their squares crowd in on 2 — but never land on it exactly. No fraction ever will.

You can keep going. You can get *astoundingly* close. But you will never, ever land exactly on 2. And the reason is not that you're not trying hard enough. The reason is that **no such fraction exists**. $\sqrt{2}$ is not a ratio of two whole numbers. It is not rational.

That word — *not rational* — gives these numbers their name. They are the **irrational numbers**. And the wild thing is we can *prove*, with total certainty, that $\sqrt{2}$ is one of them. Not “we haven't found the fraction yet.” We can prove the fraction cannot exist.

2.2 The proof (and why it's gorgeous)

You already have the fact you need: $\sqrt{2}$ is irrational, not a fraction. What follows is the *proof* of why — one of the most beautiful arguments ever made, and completely within your reach. If it gets heavy, you can skim it, trust the result, and come back later; the rest of the chapter won't depend on the details. But if you want to feel what a real mathematical proof is, slow down here.

◆ **Why it's true** — skip on a first read, but the real brain-training

The trick is sneaky. We're going to *assume the opposite of what we want to show* and watch it destroy itself. This is called **proof by contradiction**, and it's a bit like a detective assuming a suspect is innocent and then showing that assumption leads to something impossible.

So: **suppose $\sqrt{2}$ IS a fraction**. Let's see where that leads.

If it's a fraction, we can write it as $\frac{a}{b}$, where a and b are whole numbers. And we'll insist on one reasonable thing: that we've already reduced the fraction to lowest terms. (You can always do this. $\frac{6}{4}$ becomes $\frac{3}{2}$.) In particular, a and b are *not both even* — because if they were, we could cancel a 2 and reduce further. Remember that. It's the trap that's going to spring.

$$\sqrt{2} = \frac{a}{b} \implies 2 = \frac{a^2}{b^2} \implies 2b^2 = a^2.$$

Look at that last equation. The left side is 2 times something, so it is **even**. That means a^2 is even. And here's a small fact you can check yourself: if a number's square is even, the number itself must be even (an odd number squared is always odd — try a few). So a is **even**.

If a is even, we can write it as $a = 2k$ for some whole number k . Substitute that back in:

$$2b^2 = (2k)^2 = 4k^2 \implies b^2 = 2k^2.$$

But now look — the right side is 2 times something, so b^2 is **even**, which means b is **even** too.

And there's the contradiction. We just proved a is even *and* b is even. But we started by insisting they weren't both even! Our assumption has eaten itself. The only thing we assumed was that $\sqrt{2}$ could be written as a fraction. That assumption led to an impossibility. So the assumption must be false.

$\sqrt{2}$ is not a fraction. It is irrational.

There’s a legend that the man who first discovered this — a follower of Pythagoras named Hippasus — revealed the secret while at sea, and was thrown overboard for it. The Pythagoreans believed that everything in the universe was built from whole-number ratios. A number that *wasn’t* a ratio wasn’t just inconvenient. It was heresy. It meant their whole picture of reality had a hole in it.

They were right to be shaken. Because it wasn’t one hole.

2.3 The line is mostly holes

Here’s what makes this story go from “weird curiosity” to “the ground shifts under your feet.”

$\sqrt{2}$ isn’t special. The same kind of argument works for $\sqrt{3}$, $\sqrt{5}$, $\sqrt{6}$, $\sqrt{7}$ — the square root of almost any whole number is irrational. And those are just the square roots. There’s π , the number you get from a circle. There’s e , a number we’ll meet in Chapter 5. There are infinitely many more.

In fact — and this is the part that genuinely keeps mathematicians up at night — if you could somehow scoop up a number at *random* from the number line, the chance that you’d grab a nice clean fraction is **zero**. Almost every point on the line is irrational. The fractions, for all their infinite tight packing, are like scattered dust on a line that’s overwhelmingly made of irrationals.

Remember in Chapter 1, when it *felt* like the fractions must fill the whole line because you could always squeeze another one between any two? That feeling was exactly backwards. The fractions leave room for infinitely more holes than the points they actually occupy. The number line is mostly hole.

2.4 Easy to say, hard to prove

I want to stop on π for a moment, because it teaches a lesson that has nothing to do with circles.

We just proved $\sqrt{2}$ is irrational, and the whole proof fit on half a page. You could explain it to a friend over lunch. So you might reasonably expect that proving π is irrational is about the same amount of work.

It is not. The ancient Greeks could prove $\sqrt{2}$ was irrational. Nobody managed to prove it for π for another **two thousand years** — it wasn’t done until 1761, by a mathematician named Johann Lambert. And his proof needs mathematical machinery we won’t have for several books yet.

Why is one so much harder than the other? Here's the heart of it. $\sqrt{2}$ comes with a handle: it's the answer to a simple equation, $x^2 = 2$. That equation is exactly what our proof grabbed onto. π has no such handle. It is not the answer to *any* nice equation built from whole numbers. There's nothing simple to plug it into. So the proof has to come at it sideways, with much heavier tools.

How hard a fact is to *prove* has almost nothing to do with how *true* it is, or even how simple it is to state.

" π is irrational" is a short, clear sentence a child can understand. The proof is a monster. A huge amount of mathematics lives in exactly that gap — in true things that are easy to say and brutally hard to establish. When we reach calculus, you'll finally hold tools strong enough to corner π .

So we do what we always do. An operation broke — taking square roots — and the rationals couldn't handle it. So we extend. We declare that the line has *no* gaps: every single point on it is a number, fraction or not. This complete, gap-free collection — all the rationals together with all the irrationals — is called the **real numbers**.

This is the number line you've secretly been using your whole life. We just never told you it was hiding irrationals in every gap.

2.5 What does an irrational number even look like?

Fair question. You can write a fraction down exactly: $\frac{3}{8}$. How do you write $\sqrt{2}$? One way is to write it as a decimal:

$$\sqrt{2} = 1.41421356237 \dots$$

and those dots are doing serious work. The decimal goes on **forever**, and — this is the signature of an irrational number — it **never falls into a repeating pattern**.

Compare. A fraction, written as a decimal, always either stops or repeats:

$$\frac{1}{4} = 0.25 \text{ (stops), } \frac{1}{3} = 0.3333 \dots \text{ (repeats), } \frac{1}{7} = 0.\overline{142857} \text{ (repeats in sixes).}$$

That's what being a fraction *means*, written in decimals: eventually predictable. Stops, or loops. An irrational number does neither. Its decimal marches off forever with no loop, no pattern, no shortcut.

So here's a clean way to hold the whole picture in your head:

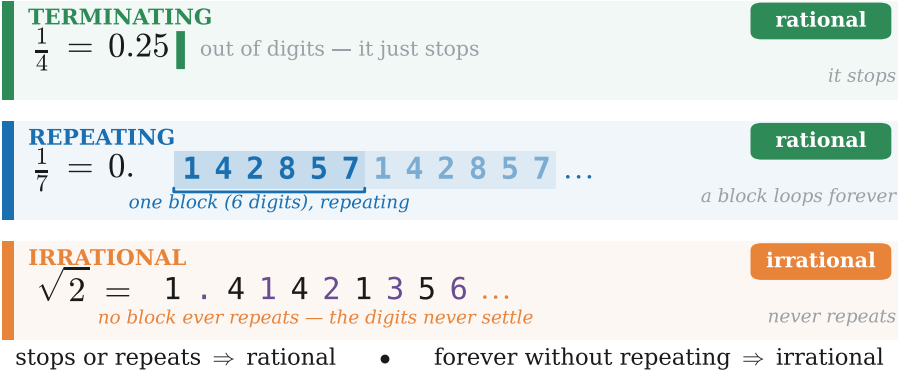


Figure 2.3. The decimal fingerprint. A decimal that *stops* or *repeats* is rational; one that runs on forever with no repeating block is irrational.

Kind of number	Its decimal
Integer	stops immediately ($5 = 5.000\dots$)
Fraction (rational)	stops, or repeats forever
Irrational	goes forever, never repeats
All of them together	the real numbers

2.6 A secret about “decimal”

Did you notice the word *decimal* keeps showing up? It comes from *decem*, Latin for ten. When you write 1.414, every digit is secretly about powers of ten. We built our whole way of writing numbers around the number **ten**.

Why ten? Honestly, because you have ten fingers. That’s it. There’s nothing mathematically special about ten — it’s an accident of human anatomy. If we’d evolved with eight fingers, we’d probably count in eights.

A computer doesn’t have fingers. Deep down, a computer is made of tiny switches, and a switch has exactly two states: **on** or **off**. So computers count in **twos**. Their natural number system uses only two digits, 0 and 1, and it’s called **binary**. In binary the places double instead of multiplying by ten: a ones place, a *twos* place, a *fours* place, an *eights* place. So the number we write as 13 becomes 1101 in binary, because

$$1101_{(\text{binary})} = 8 + 4 + 0 + 1 = 13.$$

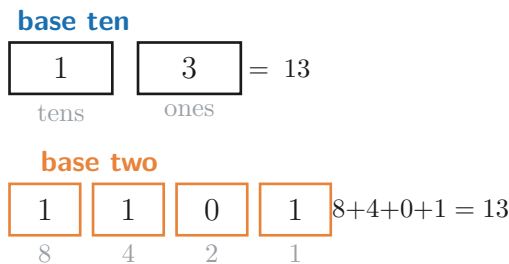


Figure 2.4. The same number, two costumes. In base ten the places go ..., 100, 10, 1; in base two they double: ..., 8, 4, 2, 1. The number 13 is 1101 in binary.

Every digit in a computer — every one of those 0s and 1s — is called a **bit**. Everything a computer has ever done is, underneath, an enormous pile of bits. The grandest machines humans have ever built are, all the way down, just patterns of *on* and *off*. Keep this in your back pocket: **ten was never sacred. It was just convenient.**

2.7 The number a computer can’t hold

Now here’s where it gets interesting, and it’s the whole reason I made you sit with infinite decimals.

To write π down *exactly*, you would need infinitely many digits. Not a lot of digits. Infinitely many. You would never finish. A computer cannot do that. It has room for maybe fifteen or sixteen digits, and then it is *out of boxes*. So when you hand it π , it writes down

3.141592653589793

and then it **stops**, and throws the rest away forever. It rounds. The number it actually stores isn't π — it's a near-miss.

And it's not just the irrationals. The fraction $\frac{1}{10}$ — plain old 0.1 — turns into a *repeating* decimal when you write it in binary, the computer's native twos. So the machine can't store 0.1 exactly either. Here's the demonstration that makes everyone's eyebrows go up. Ask almost any computer: does $0.1 + 0.2 = 0.3$? The computer says **no**. It reports the answer as 0.30000000000000004. It's not broken. It rounded 0.1 and rounded 0.2, each to the nearest dot it could store, and the two tiny roundings didn't quite cancel. A speck of error, made visible.

► **Run it yourself — watch your own computer get $0.1 + 0.2$ “wrong.”**
(notebook 2.1 in Colab)

One line of code makes your machine print 0.30000000000000004 on your own screen. Then try printing $\frac{1}{3}$ and π and see exactly where the computer runs out of room. Reading about it is a curiosity; watching your own computer do it is unforgettable.

Our number line is perfect and infinite. The computer's number line is a finite grid of dots. Most real numbers fall *between* the dots, so the computer quietly snaps each one to the nearest dot it can hold. It isn't lying to you — it's working with a very good *sketch* of the real numbers, never the real thing.

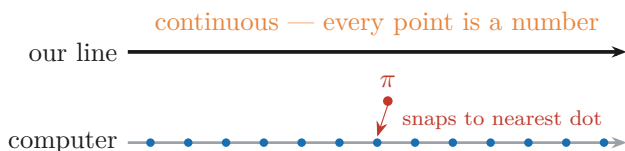


Figure 2.5. The finite grid versus the continuous line. Most reals (like π) fall *between* the computer's representable dots, so it stores the nearest dot instead — a faithful sketch, not the real thing.

Most of the time that speck of error is far too small to matter. But specks add up. When a computer trains an artificial intelligence, it performs *billions* of

these slightly-rounded steps, each feeding the next. Stack a billion tiny errors and they can grow into a real one. And notice what this quietly tells you. The careful, infinite reasoning you're learning in this book is not the clumsy version of what computers do — it's the *exact* version. **You are learning the ground truth that the machine spends all its effort trying to imitate.**

2.8 Where we are now

We started with counting. We were forced to invent negatives, then fractions, each time to repair a broken operation. And now we've been forced one more time — to fill the holes the fractions couldn't reach — and we've arrived at the **real numbers**: the complete, unbroken number line.

But notice what we've been doing. We keep talking about numbers, and then about *operations on* numbers. An operation takes a number in and gives a number out. That idea — a machine you feed a number and it hands you back another number — turns out to be even more important than the numbers themselves. It's called a **function**, and it's where we're going next.

Things worth thinking about

1. The proof that $\sqrt{2}$ is irrational hinged on one small claim: *if a number's square is even, the number itself is even*. Convince yourself by checking what happens when you square an odd number. (Try 3, 5, 7. Can you see *why* odd times odd is always odd?)
2. We proved $\sqrt{2}$ is irrational. Would the same proof work for $\sqrt{4}$? Try running the argument with 4 in place of 2 and watch where it breaks — that break is telling you something true, because $\sqrt{4}$ really *is* a nice number. What is it?
3. A fraction's decimal always stops or repeats; an irrational's never does. Here's a puzzle: I'm thinking of the decimal $0.1010010001000010\dots$, where the gaps between the 1's keep growing by one each time. It never stops. Does it repeat? Is this number rational or irrational?
4. Now you've learned there are *vastly* more irrationals than fractions. So here's the unsettling question to carry forward: are these all the same size of infinity, or are some infinities bigger than others? (They are. We'll get there.)
5. We write 13 as 1101 in binary. Work out what the binary number 10000 is in our ordinary system. (Hint: the places double — 1, 2, 4, 8, 16.) Then write the number 10 in binary.

6. A computer says $0.1 + 0.2$ is not quite 0.3 . A friend says “then computers are bad at math.” Are they right? Explain what’s actually going on — and whether the mistake is the computer’s fault or just the price of having only finitely many boxes.

Problems to work

These first three build on each other on purpose. Do them in order.

Problem 1 — Turn the crank.

We proved $\sqrt{2}$ is irrational using a fact about *even* and *odd*. Now prove that $\sqrt{3}$ is irrational. You don’t need a new idea — you need the same proof with one word swapped. Everywhere we used “even” (divisible by 2), use “divisible by 3” instead. Start: *suppose* $\sqrt{3} = \frac{a}{b}$ in lowest terms — and follow the trail until it contradicts itself. The point isn’t the answer; it’s the discovery that **the method transfers**.

★ Teach it — explain it to someone

Problem 2 — Teach it. Find a person who has never seen why $\sqrt{2}$ is irrational. *Without reading from this book*, explain it to them out loud, start to finish, until they nod. The moment you have to *say it to someone else*, the gaps in your understanding light up like warning lamps. Three load-bearing steps your explanation **must** include:

- You *assumed* $\sqrt{2}$ was a fraction in lowest terms — so a and b are not both even.
- You showed a must be even, then used that to show b must be even too.
- You pointed out that “both even” contradicts “lowest terms” — and *that’s* what kills the assumption.

Problem 3 — Stretch: adapt the idea.

Now show that $\sqrt[3]{2}$ — the cube root of 2 — is irrational. Suppose $\sqrt[3]{2} = \frac{a}{b}$ in lowest terms, cube both sides, and hunt for a contradiction. You may use this fact for free: *if 2 divides a^3 , then 2 divides a* . Push the argument through and make a and b both turn out even, contradicting “lowest terms.” You took a method built for square roots and **bent it to fit cube roots** — that is most of what doing real mathematics actually is.

Going further

★ Teach it — explain it to someone

Problem 4 — Teach it (the computer’s sketch). Sit a family member down at any computer or phone with a Python prompt, and have *them* type $0.1 + 0.2$. When the surprising 0.30000000000000004 appears, explain why — using the “finite grid of dots” picture. Get them to see the computer isn’t broken; it rounds every number to the nearest dot it can store, and two roundings didn’t quite cancel. If they walk away saying “oh — it’s a *sketch*, not the real number,” you’ve taught the whole chapter in one sentence.

Problem 5 — Build a repeating decimal back into a fraction.

A repeating decimal is secretly a fraction. Take $x = 0.\overline{27} = 0.272727\dots$. Multiply both sides by 100 to get $100x = 27.2727\dots$, then subtract the first equation from the second so the infinite tail cancels. Solve for x and write it as a fraction in lowest terms. (This proves the converse of the chapter’s claim: every repeating decimal really is rational.)

Problem 6 — [challenge] Rational, or not?

Decide whether each number is rational or irrational, and *justify* it in one line:

(a) $0.45454545\dots$ (b) $\sqrt{9}$ (c) $\sqrt{2} + \sqrt{2}$ (d) $\pi - \pi$.

(Two of these look irrational but aren’t. The lesson: “has a square root in it” does *not* mean “irrational.”)

Problem 7 — [challenge] A sum that hides.

We know $\sqrt{2}$ is irrational. Is it *possible* for an irrational number plus another irrational number to come out *rational*? Either argue it can never happen, or find two irrational numbers whose sum is a plain whole number. (Hint: think about $\sqrt{2}$ and what you’d add to it to get back to 0 ... or to 4.)

Problem 8 — [challenge] Irrational to an irrational power.

A famous, almost magical question: can an irrational number raised to an irrational power ever be *rational*? Consider $N = \sqrt{2}^{\sqrt{2}}$. You don’t know whether N is rational — and you *don’t need to*:

- If N is rational, you already have your example.
- If N is irrational, compute $N^{\sqrt{2}}$ using $(a^b)^c = a^{bc}$ — and watch it collapse to a whole number.

Either way, such a pair exists. (This is a real theorem, and this two-case trick is the actual proof. You proved something exists *without ever finding out which case is true* — a strange and powerful kind of argument.)

CHAPTER 3

Functions as Machines

At the end of the last chapter I let something slip in, almost casually, and I want to come back to it because it's the most important idea in this whole book.

We kept talking about *doing things* to numbers. Squaring. Rooting. Doubling. Each one takes a number, does something to it, and hands you back another number. Put in 5, squaring gives back 25. Put in 2, square-rooting gives back about 1.414.

That pattern — *feed in a number, get a number out* — is so ordinary that you've been doing it your whole life without naming it. But it deserves a name, because once you see it everywhere, mathematics stops being a pile of separate tricks and starts being one connected thing.

The name is **function**. And the best way to picture a function, at least at first, is as a machine.

3.1 The machine picture

Imagine a little box with a funnel on top and a spout on the bottom. You drop a number in the top. The box does something to it. A number comes out the bottom.

That's a function. This particular one we'd call the *squaring machine*. Drop in 5, out comes 25. Drop in 3, out comes 9. Drop in -4 , out comes 16 (because -4 times -4 is 16).

Mathematicians have a shorthand for this machine:

$$f(x) = x^2$$

Don't let the symbols scare you — they're just labels for the picture you already understand. Let me translate every piece:

- f is the **name** of the machine. (We could've called it Bob.)

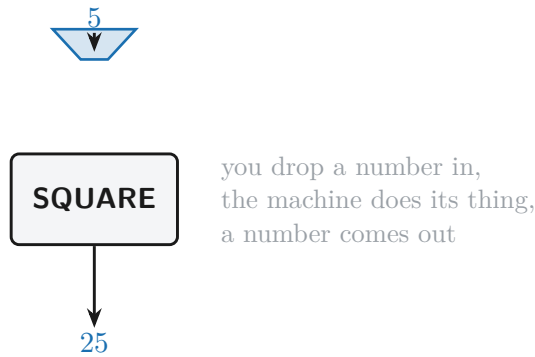


Figure 3.1. A function is a machine: input $\rightarrow$ rule $\rightarrow$ output. Drop 5 into the squaring machine and 25 comes out the spout. This box is a motif we’ll reuse all the way to neural networks.

- x is whatever number you **drop in** — a stand-in, a blank box waiting to be filled.
- $f(x)$ — read “ f of x ” — means “what the machine f gives back when you feed it x .”
- x^2 is the **rule** for what this particular machine does.

So $f(x) = x^2$ is just a compact way of writing “the machine that squares whatever you give it.” When we want to know what it does to 5, we write $f(5)$ and work it out: $f(5) = 5^2 = 25$. Once that clicks, a huge amount of mathematics suddenly becomes readable, because this notation is *everywhere*.

3.2 A machine must be reliable

Here’s a rule that seems almost too obvious to mention, but it turns out to be the heart of what makes something a function.

A function must give the same output every time you feed it the same input.

Think about why that has to be true for a *machine*. If you drop 5 into the squaring machine on Monday and get 25, then drop 5 in again on Tuesday, you’d be alarmed to get 30. A machine that gives different answers to the same question isn’t a machine you can trust — it’s broken.

This is the one unbreakable law of functions: **same input, same output. Always.** A function *is* allowed to send different inputs to the same output — the

squaring machine sends both 3 and -3 to 9, and that's fine. What it is *never* allowed to do is send one input to two different outputs. Hold onto that — in a few pages it's going to give us a clean way to look at *any* graph and instantly tell whether it's a function or not.

3.3 What can you feed it? (Domain)

Not every machine accepts every number. Take the square-root machine, $g(x) = \sqrt{x}$. Feed it 9, out comes 3. Feed it 2, out comes that irrational number from last chapter, 1.414... Fine. Now feed it -1 . The machine asks: "what number, times itself, gives -1 ?" And — at least among the real numbers — there isn't one. The machine jams.

So the square-root machine only accepts numbers from 0 upward. The collection of numbers a machine is *willing to accept* has a name: the **domain**. Figuring out a machine's domain is really just asking: **for which inputs does this machine not jam?** Two classic jams to watch for:

- **Square roots of negatives** — jams, as we saw.
- **Dividing by zero** — the machine $h(x) = \frac{1}{x}$ happily eats any number *except* 0.

3.4 What comes out? (Range)

If the domain is everything the machine will *accept*, the **range** is everything the machine can *produce*. The squaring machine is interesting here. Feed it anything you like — positive, negative, huge, tiny — and the output is *never negative*. So even though its domain is the whole number line, its range is only the numbers from 0 up.

Domain: what am I allowed to put in? **Range:** what can possibly come out?

3.5 Writing it down properly: sets and intervals

So far I've described domains in plain English. Mathematicians need to *write these down*, precisely and compactly — and so will you. A few symbols you'll see constantly:

Symbol	Read aloud as	Means
$\geq$	“greater than or equal to”	at least this big
$\leq$	“less than or equal to”	at most this big
$>, <$	“greater than”, “less than”	strictly bigger / smaller
$\in$	“is an element of”	belongs to this collection
$\mathbb{R}$	“the reals”	the whole real number line

A collection of numbers is called a **set**. There are two standard ways to write one. **Set-builder notation** describes the members by a *rule*: $\{x \mid x \geq 0\}$ reads “the set of all x such that x is greater than or equal to zero.” **Interval notation** just names the two ends: $[0, \infty)$ means “everything from 0 up to infinity.” But look closely at the brackets, because they carry real information:

- A **square bracket** $[]$ means the endpoint **is included**.
- A **parenthesis** $()$ means the endpoint is **excluded**.

And ∞ *always* gets a parenthesis, never a square bracket — because infinity isn’t a number you can reach, it’s a direction. When a machine jams at a single point in the middle, like $h(x) = \frac{1}{x}$ at 0, we glue the two surviving pieces with the **union** symbol $\cup$:

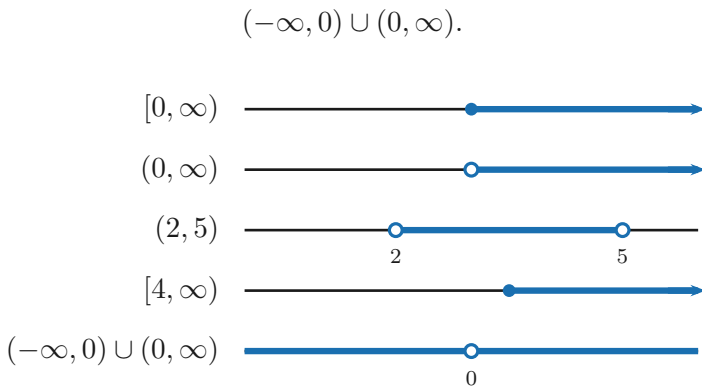


Figure 3.2. Interval notation at a glance. A filled dot (square bracket) includes the endpoint; an open dot (parenthesis) excludes it; ∞ is always open. The last row shows a domain with a hole punched out at 0.

Now we can write our machines down like grown-ups:

- Squaring $f(x) = x^2$: **domain** $(-\infty, \infty)$, **range** $[0, \infty)$.
- Square-root $g(x) = \sqrt{x}$: **domain** $[0, \infty)$, **range** $[0, \infty)$.
- Reciprocal $h(x) = \frac{1}{x}$: **domain** $(-\infty, 0) \cup (0, \infty)$, **range** the same.

3.6 Drawing the machine: graphs

There's a beautiful way to *see* an entire function at once. Take your input x and your output $f(x)$, and treat them as a *pair* — an address. The input tells you how far across to go; the output tells you how far up. Mark a dot there. Do that for every input, and the dots trace out a picture of the whole machine.

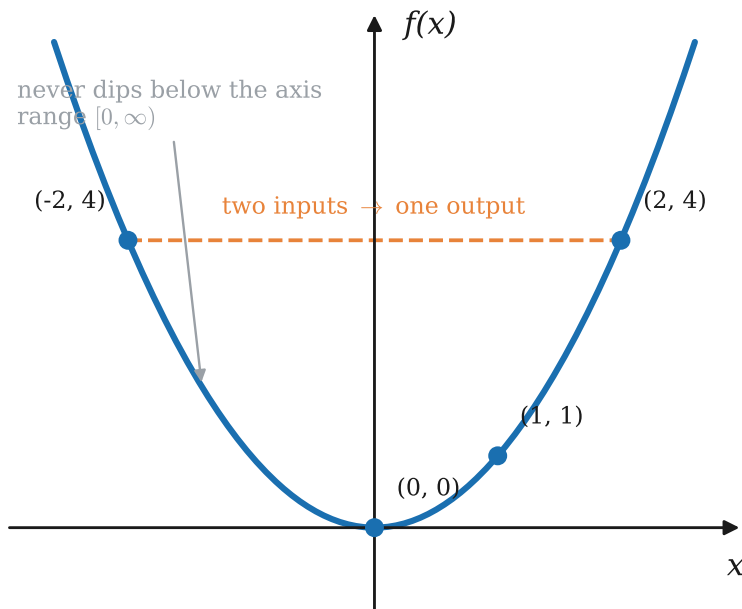


Figure 3.3. The squaring machine's graph: a *parabola*. You can read the range straight off it — the curve never dips below the axis, the visual form of “squaring never gives a negative.” Note two inputs (-2 and 2) landing on the same output (4).

The dip at the bottom sitting on 0 is the machine's smallest output. The two arms climbing upward show that big inputs — positive *or* negative — produce big positive outputs. The graph isn't decoration. It's the whole machine, laid out flat so your eyes can take it in at a glance.

3.7 The trick that tells you “yes, this is a function”

Remember the one unbreakable law? *Same input, same output* — one input can never give two different outputs. On a graph, the horizontal direction is the input and the vertical is the output. So “one input gives two different outputs” would look like a single spot on the horizontal axis having *two dots stacked above it*. Which gives us a dead-simple test:

The vertical line test. Sweep a vertical line across the graph. If it ever touches the graph in two places at once, it's *not* a function — that single input is trying to produce two outputs.

The vertical line test = “same input, same output,” seen sideways

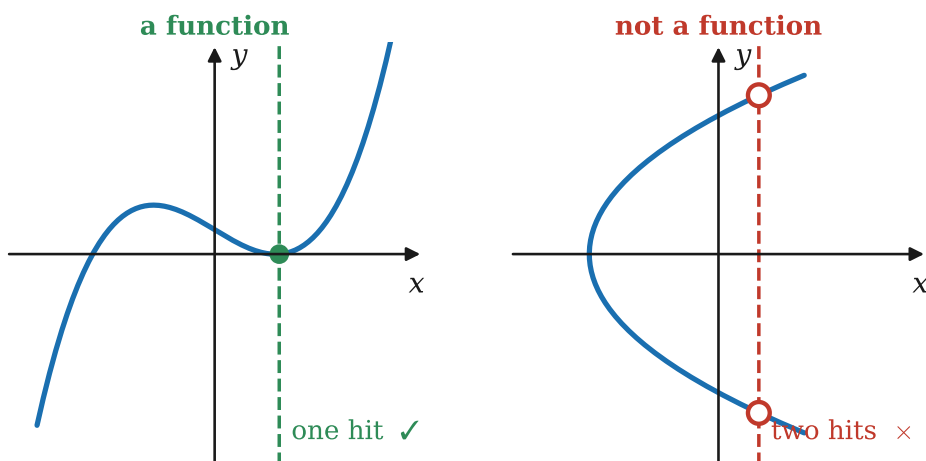


Figure 3.4. The vertical line test is just the “same input, same output” law seen sideways. The parabola passes (every vertical line hits once); the sideways parabola fails (a vertical line hits twice).

It isn't a separate rule to memorize. It's the “same input, same output” law *seen sideways*. That's the payoff of understanding *why* the law exists — the test stops being mysterious and becomes something you could have invented yourself.

3.8 Chaining machines together (composition)

Here's where functions start to feel powerful. What if you took the output of one machine and fed it straight into another? Bolt the spout of the first onto the funnel of the second. Now you've built a *bigger* machine out of two smaller ones.

Say machine one *adds 1*: $g(x) = x + 1$. And machine two *squares*: $f(x) = x^2$. Drop in 3. The first machine makes it 4. The second squares that into 16. We write this chaining as $f(g(x))$ — read “ f of g of x ” — and you read it *from the inside out*, just like the order the number actually travels. This is called **composition**, and it is, quietly, one of the biggest ideas you'll ever meet.

Why so big? Because *everything complicated is made of simple things chained together*. A long, intimidating formula is almost always just a handful of tiny, boring machines bolted end to end. And when an artificial intelligence “thinks,”

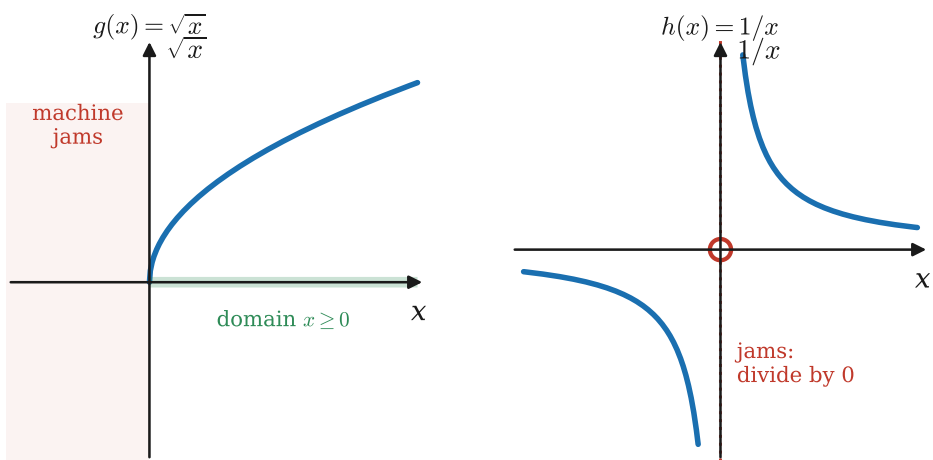


Figure 3.5. Two classic domain jams: $\sqrt{x}$ refuses negative inputs (it jams for $x < 0$), and $1/x$ jams at $x = 0$ where you’d divide by zero. The domain is exactly the set of inputs that *don’t* jam the machine.

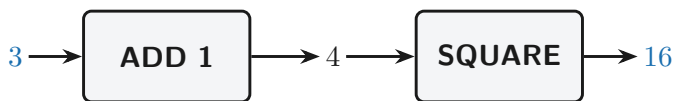


Figure 3.6. Composition: two machines in series. The intermediate value 4 travels from one spout to the next funnel. We write this $f(g(x))$ and read it *inside-out*. (This is the future neural-network motif — keep the visual in mind.)

what it is really doing is passing a number through a staggeringly long chain of simple machines, one feeding the next, thousands deep.

One warning: **order matters**. Squaring-then-adding-1 is a *different* machine from adding-1-then-squaring. Try 3 the other way: square first to get 9, then add 1 to get 10 — not 16. Exactly like putting on socks-then-shoes is not the same as shoes-then-socks.

3.9 Running the machine backwards (inverses)

Some machines can be *undone*. The *add-5* machine takes 3 to 8. Is there a machine that takes 8 back to 3? Of course — the *subtract-5* machine. We say subtract-5 is the **inverse** of add-5: the “undo” button.

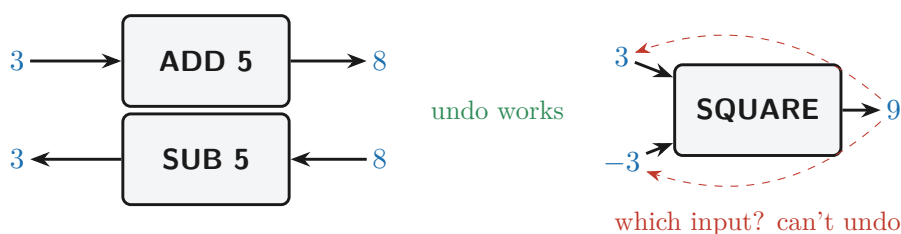


Figure 3.7. An inverse is an undo button. Add-5 and subtract-5 reverse each other perfectly. But squaring sends both 3 and -3 to 9 — so running it backwards from 9 can’t decide which input to return. It threw information away.

But here’s the catch. Look again at the squaring machine. It sends 3 to 9. It *also* sends -3 to 9. So if I tell you the output was 9 and ask you to run the machine backwards, you’re stuck. Was the original input 3, or -3 ? You genuinely cannot tell. The machine threw away a piece of information (the sign), and no undo button can recover what’s been thrown away.

A machine can be run backwards only if it never sends two different inputs to the same output. The moment two inputs collide into one output, the machine has destroyed the information needed to reverse it.

Notice that’s the *opposite* worry from the vertical line test. There we asked “does one input give two outputs?” (never allowed). Here we ask “do two inputs give one output?” (allowed in general, but exactly what blocks an inverse). Two different questions — and you now understand both from first principles.

3.10 Where we are now

We started with a single picture — a machine with a funnel and a spout — and pulled almost everything from it:

- a **function** is a reliable machine: same input, same output;
- the **domain** is what it will accept, the **range** what it can produce — both writable in set-builder or interval notation;
- the **graph** is the whole machine laid flat;
- the **vertical line test** is just the one law seen sideways;
- **composition** chains simple machines into complicated ones — the secret behind everything from long formulas to artificial intelligence;
- an **inverse** is an undo button, which exists only when the machine hasn't thrown information away.

Everything from here on is about *specific* machines — polynomials, exponentials and logarithms, trig functions, and finally the strange numbers that let the square-root machine accept negatives at last. They're not a zoo of formulas to memorize. They're just particular machines, and you already understand what a machine *is*. That's the hard part, and you've done it.

Things worth thinking about

1. The squaring machine sends both 3 and -3 to 9. Can you think of another machine that sends two *different* inputs to the *same* output? Now can you think of a machine that *never* does this — where every output comes from exactly one input?
2. We said order matters in composition: add-1-then-square gave 16, but square-then-add-1 gave 10. Find *two* machines where the order *doesn't* matter. (They're rarer than you'd think. What's special about the ones you found?)
3. The graph of a full circle fails the vertical line test, so a circle isn't a function. But the *top half* of a circle passes. Why? And what does that tell you about cutting a “non-function” into pieces that *are* functions?
4. (*Looking ahead.*) The squaring machine can't be undone because it can't decide between 3 and -3 . But what if we only ever allowed positive inputs — what if we *narrowed the domain* so the machine never sees both? Could it be undone then? (This little trick turns out to matter a lot later.)

Problems to work

Problem 1 — Read the machine, and write it down properly.

For each machine, work out its **domain** and **range**, in **interval notation**. Say out loud what would make each one *jam*, then translate that into brackets.

$$(a) f(x) = x + 7 \quad (b) g(x) = \frac{1}{x-2} \quad (c) h(x) = \sqrt{x-4}$$

(For (b): what input jams it? Your domain will be a union of two pieces. For (c): the thing under the root can't go negative — so how small can x be, and does that smallest value get a square bracket or a parenthesis?)

Problem 2 — Chain them, both ways.

Let $f(x) = 2x$ and $g(x) = x + 3$. (a) Compute $f(g(5))$ — add 3 first, *then* double. (b) Compute $g(f(5))$ — double first, *then* add 3. (c) You get two different answers. Write one sentence explaining *why* the order changed the result.

★ Teach it — explain it to someone

Problem 3 — Teach it (the inverse idea). Find someone who hasn't thought about this, and pose them a puzzle: “*I have a machine. I put in a secret number, and out came 9. The machine squares its input. What was my secret number?*” Let them sit with it until they notice the trap — that it could have been **3 or** -3 , with no way to be sure. Then explain why this is exactly the reason the squaring machine has no undo button. If you can make someone else *feel* the ambiguity, you understand inverses better than a definition could ever give you.

Going further

★ Teach it — explain it to someone

Problem 4 — Teach it (domain jams). Pick the machine $g(x) = \frac{1}{x-2}$ and explain to a family member, in plain words, why you're *allowed* to put in $x = 100$ or $x = -7$ but *not* $x = 2$. Have them try $x = 2$ on a calculator and watch it complain. The idea to land: a domain isn't a rule someone made up — it's just “the inputs that don't break the machine.”

Problem 5 — Evaluate carefully.

For $f(x) = x^2 - 3x + 1$, compute $f(0)$, $f(2)$, $f(-1)$, and $f(\frac{1}{2})$. (Substitute, *then* evaluate — every time. Watch the signs on $f(-1)$.)

Problem 6 — Match the graph to its rule.

Without plotting carefully, decide which of these can possibly be the graph of a function of x , using the vertical line test in your head: (a) a straight vertical line; (b) a horizontal line; (c) a “V” shape; (d) a full circle; (e) the letter “S” lying on its side. For each “no,” name one input that would have two outputs.

Problem 7 — [challenge] Build the inverse.

The machine $f(x) = 2x + 6$ can be undone. Find its inverse machine f^{-1} — the one that takes any output back to the input that produced it. (Hint: f *doubles then adds 6*; undo those two steps *in reverse order* — subtract 6 first, then halve. Write $f^{-1}(x)$ as a formula, then test it: feed $f(5)$ into f^{-1} and confirm you get 5 back.)

Problem 8 — [challenge] When does narrowing the domain rescue an inverse?

The squaring machine has no inverse because 3 and -3 collide on 9. Suppose we *forbid* negative inputs — restrict the domain to $[0, \infty)$. (a) Explain why the machine now *can* be undone. (b) What is the undo machine? (c) Why is this exactly why $\sqrt{}$ is defined to give only the *positive* root? (This is a preview of a trick used constantly with trig functions later.)

Problem 9 — [challenge] Compose three.

Let $a(x) = x + 1$, $b(x) = 2x$, and $c(x) = x^2$. Work out $a(b(c(2)))$ by travelling inside-out (do c first). Then compute $c(b(a(2)))$ and confirm it’s different. Finally, describe in words the single combined machine $a(b(c(x)))$ as a formula in x . (This is how a three-layer “network” processes a number — you’re hand-running a tiny one.)

CHAPTER 4

Polynomials, Factoring, and the Quadratic Formula

You now know what a function *is* — a reliable machine, with a domain and a range, that you can graph, chain, and sometimes reverse. From here on we meet the *famous* machines, the ones that show up so often they’ve earned names. This chapter is about the simplest and friendliest family of all.

They’re so simple you can build every one of them using nothing but the operations you’ve trusted since Chapter 1: adding, subtracting, and multiplying. No division, no roots, no exotic ingredients. These are the **polynomials**.

4.1 What a polynomial is

A polynomial is any machine you can write as a sum of terms, where each term is a number times x raised to a whole-number power. That sounds stuffy, so look at one:

$$f(x) = 3x^2 + 2x - 5$$

Three terms. The first is 3 times x^2 . The second is 2 times x (which is x^1). The last is -5 , which you can think of as $-5 \times x^0$, since anything to the power 0 is 1. Every term is *a number times a whole-number power of x* . That’s the whole definition.

And here are some things that are **not** polynomials, so you can feel the boundary: $\sqrt{x}$ (that’s x to the power $\frac{1}{2}$ — not a whole number), $\frac{1}{x}$ (power -1 — negative), and 2^x (the x is up in the *exponent* — a completely different beast, the next chapter’s).

Because they’re built only from multiplying and adding, polynomials never jam:

Every polynomial accepts every real number. Its domain is always $(-\infty, \infty)$.

4.2 The vocabulary (just two words)

The **degree** of a polynomial is its highest power. $3x^2 + 2x - 5$ has degree 2; $x^3 - 2x^2 + x - 8$ has degree 3; plain 7 has degree 0. Degree controls the machine's whole personality. The **leading term** is the term with that highest power. When x gets huge, the leading term bullies all the others into irrelevance, so it alone decides what the machine does far out at the edges. Degrees have nicknames: degree 1 is **linear**, degree 2 is **quadratic**, degree 3 is **cubic**.

4.3 Roots: where the machine hits zero

Here's the single most useful question you can ask about a polynomial:

For what inputs does the machine output exactly zero?

Those special inputs are called the **roots** (or *zeros*). On a graph, they're exactly where the curve *crosses or touches the horizontal axis*.

Why care so much? Because an astonishing amount of mathematics is secretly the question “*when does this equal that?*” — and you can always rearrange “this = that” into “this – that = 0.” So “when are two things equal?” becomes a question about roots. Finding roots is the engine under a huge fraction of all the algebra you'll ever do.

4.4 Factoring: breaking a machine into pieces

Here's a fact that looks small and turns out to be a superpower.

If two things multiply to give zero, then at least one of them must have been zero. If $A \times B = 0$, then $A = 0$ or $B = 0$ (or both).

Now suppose I hand you a quadratic already written as a *product*: $f(x) = (x - 2)(x - 3)$. When is it zero? Exactly when $x - 2 = 0$ (so $x = 2$) or $x - 3 = 0$ (so $x = 3$). The roots are 2 and 3, read *straight off the pieces*.

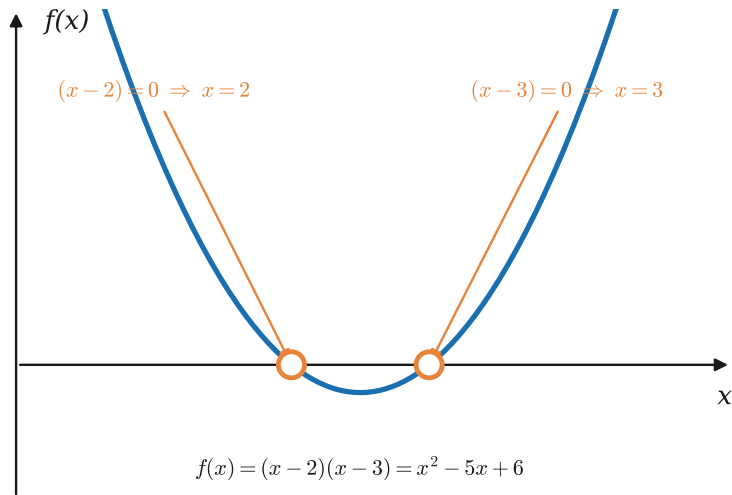


Figure 4.1. Factoring as decomposition. Written as $(x-2)(x-3)$, the curve's roots fall out for free: it crosses zero exactly where each factor vanishes, at $x=2$ and $x=3$.

The catch is that polynomials usually arrive multiplied out: $x^2 - 5x + 6$. This is the *same machine* as $(x-2)(x-3)$, but in this form the roots are hidden. **Factoring** is running that multiplication backwards. For $x^2 - 5x + 6$, factoring means finding two numbers that *multiply* to $+6$ and *add* to -5 : those are -2 and -3 , so it factors as $(x-2)(x-3)$, and the roots are 2 and 3 .

4.5 A factoring toolkit you should own

Some factorings come up so often you should *know* them on sight. We'll build the toolkit by *multiplying out*, because watching the expansion is the most honest proof that the identity is true.

1. Difference of squares. The most useful identity in elementary algebra:

$$a^2 - b^2 = (a - b)(a + b).$$

Multiply the right side out: $(a - b)(a + b) = a^2 + ab - ba - b^2$. The two middle terms cancel, leaving $a^2 - b^2$. So $x^2 - 9 = (x - 3)(x + 3)$, instantly.

2. Perfect-square trinomials. What you get when you square a sum or difference:

$$a^2 + 2ab + b^2 = (a + b)^2 \quad a^2 - 2ab + b^2 = (a - b)^2.$$

This time the middle terms *add*, giving the doubled $2ab$. **Recognizing this**

pattern is exactly what “completing the square” is about — the trick we’re about to use to crack every quadratic.

3. Sum and difference of cubes. Less famous, equally worth knowing:

$$a^3 - b^3 = (a - b)(a^2 + ab + b^2), \quad a^3 + b^3 = (a + b)(a^2 - ab + b^2).$$

4. The grand pattern: $x^n - 1$. This one holds for *every* whole-number power n :

$$x^n - 1 = (x - 1)(x^{n-1} + x^{n-2} + \cdots + x + 1).$$

In words: $x^n - 1$ always has $(x - 1)$ as a factor, and the other factor is *every power of x from 0 up to $n - 1$, all added together*.

◆ **Why it’s true** — skip on a first read, but the real brain-training

Here’s the proof, and it uses a trick worth stealing. Multiply $(x - 1)$ through the long sum. The x hits every term, bumping each power up by one; the -1 copies each term unchanged with a minus sign:

$$\begin{aligned} x \cdot (x^{n-1} + \cdots + x + 1) &= x^n + x^{n-1} + \cdots + x^2 + x \\ -1 \cdot (x^{n-1} + \cdots + x + 1) &= -x^{n-1} - \cdots - x^2 - x - 1 \end{aligned}$$

Now add the two rows. Look down the columns: x^{n-1} cancels $-x^{n-1}$, x^2 cancels $-x^2$, x cancels $-x$ — *every term in the middle annihilates its twin*, like a row of dominoes knocking each other flat. This is a **telescoping** sum, because the whole thing collapses like a folding telescope. The only survivors are the very first term, x^n , and the very last, -1 , giving $x^n - 1$.
✓

And notice — difference of squares is just the $n = 2$ case of this very pattern. Rearrange it slightly — divide both sides by $(x - 1)$ — and it becomes the formula for a **geometric sum**:

$$1 + x + x^2 + \cdots + x^{n-1} = \frac{x^n - 1}{x - 1}.$$

Keep this one. You will meet it at least three more times: adding up any geometric chain (compound interest, a bouncing ball, a repeating decimal); probability (“how many tries until the first success?”); and complex numbers, where $x^n = 1$ has n solutions arranged as the corners of a polygon on a circle — the *roots of unity*. Same algebra, four different worlds.

4.6 A triangle hiding in the algebra

Before we crack the quadratic, here's something delightful — no work required, just notice it. What if we take higher powers of a sum? Line up the results:

$$(a + b)^1 = 1a + 1b$$

$$(a + b)^2 = 1a^2 + 2ab + 1b^2$$

$$(a + b)^3 = 1a^3 + 3a^2b + 3ab^2 + 1b^3$$

$$(a + b)^4 = 1a^4 + 4a^3b + 6a^2b^2 + 4ab^3 + 1b^4$$

Stare at just the *numbers in front* — the coefficients. That's a triangle, and it has a magic building rule: **every number is the sum of the two just above it**.

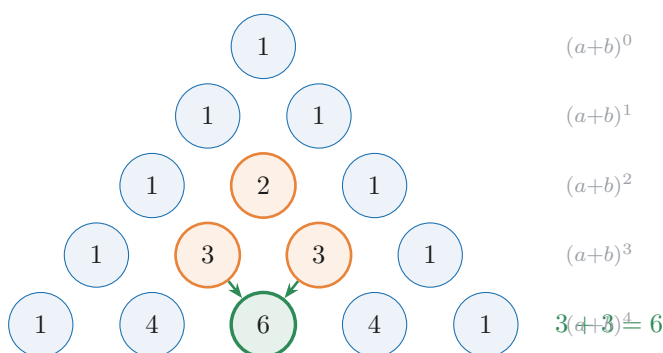


Figure 4.2. Yang Hui's triangle. Each entry is the sum of the two above it (here $3 + 3 = 6$), and row n gives the coefficients of $(a + b)^n$. Extend it forever with no multiplying — the next row is 1, 5, 10, 10, 5, 1.

This is **Yang Hui's triangle** — named for the Chinese mathematician who wrote it down in 1261, though it was known to Jia Xian earlier and appears in Persian and Indian mathematics too. (In many Western books it's called *Pascal's triangle*, after a European who studied it four centuries later.) When you reach the probability book, these exact numbers will walk back through the door — because the number of ways to flip a coin and get a certain number of heads *is* a number in this triangle. For now, just enjoy the pattern.

4.7 When factoring won't cooperate: build your own formula

Factoring by hand works beautifully when the numbers are friendly. But plenty of quadratics have roots that *aren't* tidy, and then the guess-two-numbers game just fails. We need something that works *every* time — and we're going to *build* it, so you never have to take it on faith.

The one idea. Square roots undo squares. If I can force a quadratic into the shape $(\text{something})^2 = \text{a number}$, then I can square-root both sides and I'm basically done. The bending move is called **completing the square**, and the name is literal.

Solve $x^2 + 6x + 5 = 0$. **Step 1: get the loose number out of the way.** Move the $+5$ across: $x^2 + 6x = -5$. **Step 2: complete the square.** A perfect square $(x + p)^2$ expands to $x^2 + 2px + p^2$. Matching the middle: $2p = 6$, so $p = 3$, and the missing piece is $p^2 = 9$. Add 9 to *both* sides:

$$x^2 + 6x + 9 = -5 + 9 \implies (x + 3)^2 = 4.$$

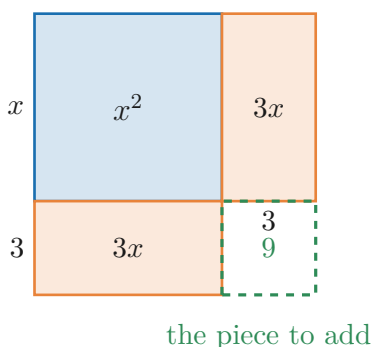


Figure 4.3. Completing the square, literally. The x^2 square plus two 3-wide strips ($3x$ each) leave a missing $3 \times 3 = 9$ corner. Adding that 9 *completes* the square into $(x + 3)^2$.

The whole left side collapsed into a single square. **Step 3: undo the square** — and a square root has *two* answers, so we write $\pm$: $x + 3 = \pm 2$. **Step 4: solve.** $x = -3 \pm 2$, which means $x = -1$ or $x = -5$. Notice where the *two* roots came from — the $\pm$, which came from the fact that square roots have two answers.

♦ **Why it's true** — skip on a first read, but the real brain-training

Here is the result, the **quadratic formula**, which solves *every* quadratic $ax^2 + bx + c = 0$ at once:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}.$$

But where does it come from? It's nothing more than the completing-the-square you just did, run once on letters instead of numbers. Divide through by a : $x^2 + \frac{b}{a}x + \frac{c}{a} = 0$. Move the loose number across: $x^2 + \frac{b}{a}x = -\frac{c}{a}$.

Complete the square — half the middle coefficient is $\frac{b}{2a}$, squared is $\frac{b^2}{4a^2}$ — and add it to both sides:

$$\left(x + \frac{b}{2a}\right)^2 = \frac{b^2 - 4ac}{4a^2}.$$

Undo the square with a $\pm$, then move the $\frac{b}{2a}$ across:

$$x + \frac{b}{2a} = \pm \frac{\sqrt{b^2 - 4ac}}{2a} \implies x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}.$$

There it is — just completing the square, done once and for all, so nobody ever has to do the four steps again.

A formula is **frozen thought**. You do the hard reasoning *once*, in general, with letters — and from then on it's reusable. You can *remember* the quadratic formula, or, if you forget it, *rebuild* it from the one idea. The second skill is the one that actually makes you a mathematician.

4.8 The discriminant: a fork in the road

Look hard at the one piece sitting under the square root: $b^2 - 4ac$. This little expression has its own name — the **discriminant** — because it *discriminates*, in the old sense of “tells things apart.” Its sign alone tells you what kind of roots you'll get:

- $b^2 - 4ac > 0$: two *different* real roots. The graph crosses the axis twice.
- $b^2 - 4ac = 0$: one repeated root. The graph just *kisses* the axis and turns back.
- $b^2 - 4ac < 0$: you're asked for the square root of a *negative number*.

Stop right there. We are at the exact spot I promised back in Chapter 1. The square-root machine *jams* on negative inputs. So when the discriminant comes out negative, the formula is demanding a number that, among the reals, simply does not exist. For two and a half thousand years, this is where mathematicians stopped. But you know this story by now. Every time an operation has jammed, we *invented the numbers that made it work*.

The discriminant decides how many times the curve meets the axis

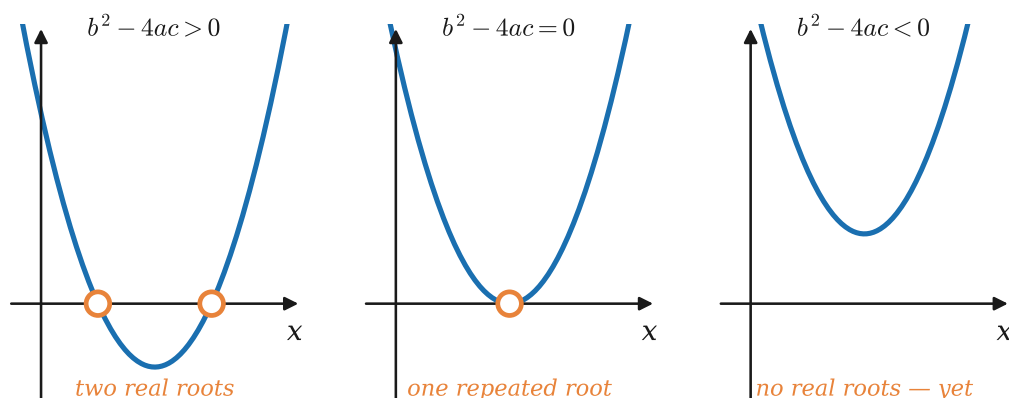


Figure 4.4. The discriminant fork. The sign of $b^2 - 4ac$ decides whether the parabola crosses the axis twice, kisses it once, or floats clear of it (no real roots — yet).

The negative discriminant is the door, and in Chapter 8 we walk through it. When we do, every quadratic — *every single one* — will turn out to have exactly two roots.

4.9 A glimpse of deep water

If you’ve been paying attention, a question should be itching at you. We have a formula for the *quadratic*. So what about degree 3? Is there a **cubic formula**? And a quintic, and a quintic, forever?

Degrees 3 and 4: yes, but at a terrible price. There *is* a cubic formula, found in Renaissance Italy in the 1500s amid a soap opera of broken oaths and public duels. Tartaglia worked out how to solve cubics and told Cardano under a sworn vow of secrecy — which Cardano broke, publishing it in 1545. His student Ferrari cracked the *quartic* too. The cubic and quartic formulas are real — but they are *monsters*, pages of nested roots, almost useless by hand.

Degree 5: no. And not “we couldn’t find it” — “it is impossible.” In the early 1800s a young Norwegian, Niels Henrik Abel, proved there is *no* formula for the general quintic. Not a complicated one — *none can exist*, ever, built from $+$, $-$, $\times$, $\div$ and roots. The pattern that held for degrees 2, 3, 4 simply *shatters* at 5.

Why does it break? Answering that required a French teenager, Évariste

Galois, to invent an entirely new kind of mathematics — not about numbers at all, but about **symmetry**: the ways the roots can be shuffled without anything noticing. His tool is now called **group theory**, and groups turned out to describe symmetry *everywhere* — in crystals, in physics, in the codes that keep your messages private. Galois wrote down the heart of these ideas the night before he was killed in a duel, at the age of twenty.

The point of telling you now: **mathematics is far bigger than the formulas in this book**. Underneath the everyday machinery there are vast, strange, beautiful structures that often look “useless” when discovered and turn out to hold up the modern world.

4.10 The shape of a polynomial

A quick word on what these machines *look* like. The degree controls the basic shape: degree 1 is a straight line (one crossing); degree 2 a parabola (up to two); degree 3 an S-like curve (up to three).

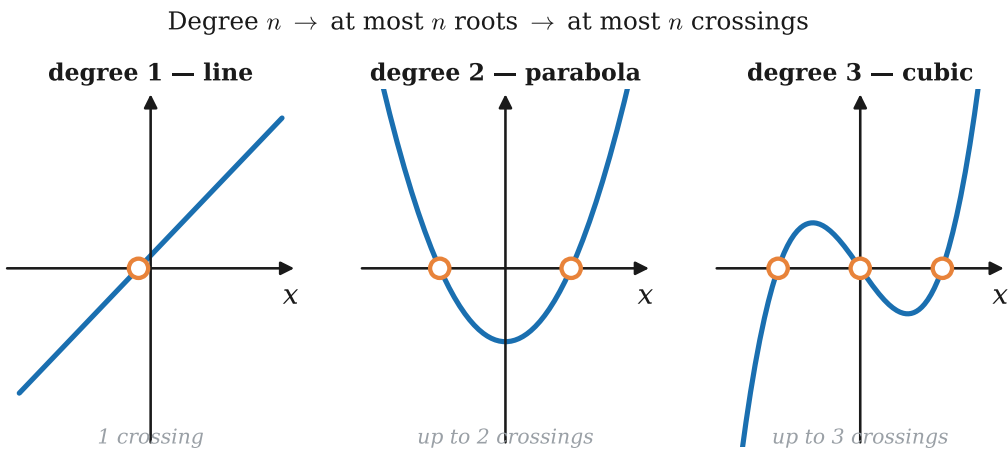


Figure 4.5. Polynomial shapes by degree. A degree- n polynomial can cross the axis at most n times — the highest power is a hard ceiling on the number of roots.

A polynomial of degree n can have at most n roots.

This is your first glimpse of the *Fundamental Theorem of Algebra* — which, once we have complex numbers, will sharpen “at most n ” into “*exactly* n , if you count them right.” Another reason Chapter 8 matters.

The pattern holds... then shatters at 5

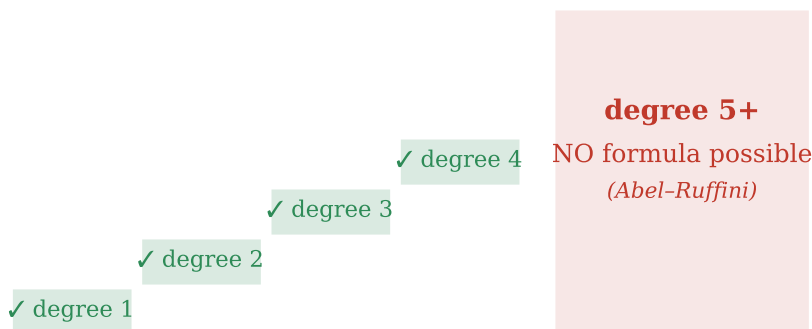


Figure 4.6. The solvability ladder. Degrees 1 through 4 each have a formula built from roots; at degree 5 the pattern hits a wall and never recovers (Abel–Ruffini).

4.11 Where we are now

Polynomials are the friendly family: built from only adding and multiplying, accepting every real number, never jamming. We learned to read their **degree** and **leading term**, to hunt for their **roots**, to **factor** them when the numbers cooperate, and to fall back on the **quadratic formula** when they don’t. And right at the heart of that formula we found the **discriminant**, whose negative branch marches us straight back to the oldest pattern in the book: an operation has jammed, and a new kind of number is waiting to be invented. Next we meet the machines that *grow*.

Things worth thinking about

1. Plain 7 is a polynomial of degree 0 — a machine that ignores its input. What does its graph look like? How many roots does it have? (Careful — when is 7 ever equal to zero?)
2. We said a degree- n polynomial has *at most* n roots. Imagine a parabola with exactly two roots, then one with exactly one, then one with zero. What’s physically different about where the three sit relative to the axis?
3. The discriminant tells you how many real roots a quadratic has *before* you finish solving it. When would you want to know “how many roots” without needing the roots themselves?
4. (*Looking ahead.*) When the discriminant is negative, the parabola “never

touches the axis.” Yet we claim it secretly has two roots. How can a curve have roots it never visibly touches? Let the strangeness sit — that discomfort is a new kind of number about to be born.

5. (*Deep water.*) We can solve degrees 1–4 with formulas, but degree 5 has *none* — and that was *proven*. What’s the difference between “nobody has found it yet” and “it’s impossible and here’s why”? Can you think of anything else proven impossible, rather than merely never done?

Problems to work

Problem 1 — Factor and read the roots.

Each factors into nice whole-number pieces. Find two numbers that *multiply* to the constant and *add* to the middle coefficient, write the factored form, and read off the roots.

$$(a) \ x^2 - 7x + 12 \qquad (b) \ x^2 + x - 6 \qquad (c) \ x^2 - 9$$

(Sneaky (c): what two numbers multiply to -9 and add to 0 ?)

Problem 2 — Complete the square yourself.

Solve $x^2 + 8x + 7 = 0$ using the four steps from the chapter — *don’t* use the finished formula. Get the loose number across, find the piece that completes the square, add it to both sides, fold into (something)², square-root with a $\pm$, and solve. Then check by factoring.

Problem 3 — When factoring fails, turn the crank.

The quadratic $x^2 - 4x + 1 = 0$ does *not* factor nicely — try for a minute and you’ll see. Now use the quadratic formula. Identify a , b , c , plug in, and find both roots. (Your answers will involve a square root that doesn’t simplify — that’s expected, and exactly why factoring failed.)

Problem 4 — Read the fork.

For each quadratic, compute *only* the discriminant $b^2 - 4ac$ and use its sign to say — without finding the roots — whether it has two real roots, one repeated, or none.

$$(a) \ x^2 - 6x + 9 \qquad (b) \ x^2 + 2x + 5 \qquad (c) \ 2x^2 + 3x - 1$$

When you finish (b), pause: you’ve found a quadratic with no real roots. Keep it in your pocket — Chapter 8 will give it the two roots it’s hiding.

★ Teach it — explain it to someone

Problem 5 — Teach it. Explain to someone why $(x - 2)(x - 3) = 0$ forces x to be either 2 or 3 — and why the same trick would *not* let you solve $(x - 2)(x - 3) = 12$ by setting each piece equal to 12. (This is a mistake almost everyone makes once. Getting someone else to see *why* it's wrong will lock the zero-product idea into your own head for good.)

Problem 6 — Verify a toolkit identity.

Multiply out the right side of $(a - b)(a^2 + ab + b^2)$ term by term, and watch the middle terms cancel until only $a^3 - b^3$ survives. (Hint: you'll get six terms before any cancelling — write them all down first, *then* hunt for the pairs that kill each other.)

Problem 7 — Use the toolkit.

Factor each *on sight*, using a tool from the toolkit. Name which tool you used. (a) $x^2 - 49$; (b) $x^2 + 10x + 25$; (c) $x^3 - 8$ (hint: $8 = 2^3$, a difference of cubes).

Problem 8 — Extend the triangle.

Write the next *two* rows of Yang Hui's triangle below 1, 4, 6, 4, 1, using only the build rule. Then use your new row to write down $(a + b)^5$ without multiplying anything.

Going further

★ Teach it — explain it to someone

Problem 9 — Teach it (frozen thought). Explain to a parent or sibling why the quadratic formula isn't something to be afraid of — it's just completing-the-square “done once, with letters, so nobody has to do it again.” Walk them through the easy numeric example $x^2 + 6x + 5 = 0$ first, then point at the general version and show it's the *same four moves*. The idea to land: a formula is *frozen reasoning*, and if you ever forget it you can thaw it back out.

Problem 10 — [challenge] A disguised difference of squares.

Factor $x^4 - 16$ completely. (Hint: it's a difference of squares — $x^4 = (x^2)^2$ and $16 = 4^2$ — but one of the pieces you get is *itself* a difference of squares. Keep going until nothing factors further over the reals. One factor will stubbornly refuse: $x^2 + 4$. Remember it for Chapter 8.)

Problem 11 — [challenge] Build a quadratic to order.

Find a quadratic $x^2 + bx + c$ whose two roots are exactly 5 and -2 . (Hint: work *backwards* — if the roots are 5 and -2 , what must the factored form be? Multiply it out.) Then do it again for roots 3 and 3 (a repeated root), and check that your second quadratic has discriminant zero.

Problem 12 — [challenge] Sum and product of roots.

For $x^2 + bx + c = (x - r)(x - s)$, multiply the right side out and match coefficients to show that $r + s = -b$ and $rs = c$. (So the two roots always *add* to $-b$ and *multiply* to c — which is exactly why the “find two numbers that add and multiply” factoring game works!) Use this to find, in your head, two numbers that add to 7 and multiply to 10 — and hence factor $x^2 - 7x + 10$.

Problem 13 — [challenge] The geometric-sum payoff.

Use the identity $1 + x + x^2 + \cdots + x^{n-1} = \frac{x^n - 1}{x - 1}$ to add up $1 + 2 + 4 + 8 + \cdots + 1024$ *without* adding the eleven numbers one by one. (Hint: this is the doubling chain, so $x = 2$, and the last term $1024 = 2^{10}$ tells you n .) Then marvel: a chessboard’s 64 doubling squares total $2^{64} - 1$ grains — more rice than humanity has ever grown.

CHAPTER 5

Exponentials and Logarithms

Back in Chapter 2 I wrote down the most natural list of numbers in the world:

$$1, \quad 2, \quad 4, \quad 8, \quad 16, \quad 32, \quad \dots$$

Each one is double the one before. A folded sheet of paper, a splitting cell, a rumor passed to two new people each day — doubling is everywhere.

In Chapter 4 we met the polynomial machines, where x sits in the *base* and a fixed number rides on top: x^2 , x^3 . This chapter is about what happens when you swap those roles — nail the base down to a fixed number and let x climb up into the *exponent* instead. That one swap produces a completely different machine, and it governs almost everything that grows or fades in the real world: money, populations, radioactivity, epidemics, the cooling of your coffee. These are the **exponential** machines, and their mirror-image undo buttons, the **logarithms**.

5.1 From doubling to a machine

Label each number in the doubling list by *how many doublings* it took to reach it, starting from 1:

doublings	value
0	1
1	2
2	4
3	8
4	16

The right column is just powers of 2: $2^0, 2^1, 2^2, 2^3, 2^4$. So “the number after x doublings” is a *machine*: feed in x , get out 2^x . We write it $f(x) = 2^x$ and call it an **exponential function** with **base 2**.

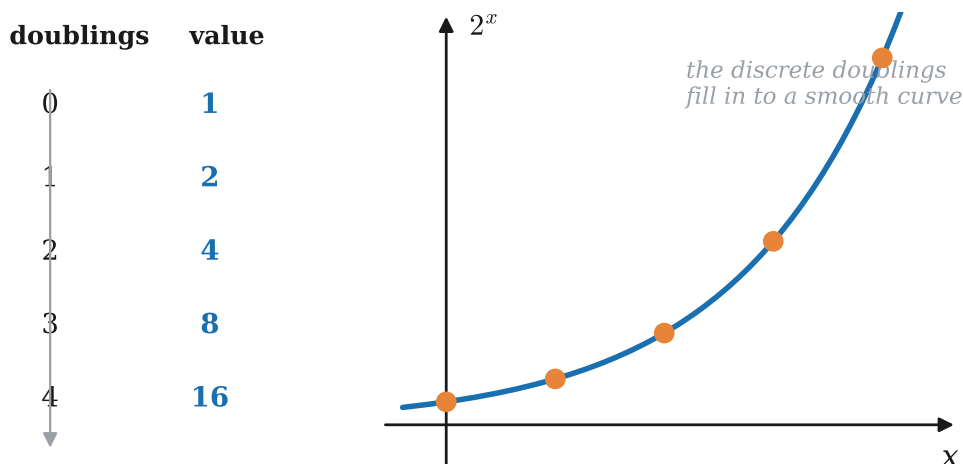


Figure 5.1. The discrete doubling sequence fills in to a continuous curve. The dots are the whole-number doublings; the smooth curve 2^x joins them up for every real input in between.

Notice how different this is from the polynomials. In x^2 , the input is the base. In 2^x , the input has moved up into the exponent. That’s the defining feature of an exponential: **the variable is in the exponent**.

5.2 What does 2^x mean when x isn’t a whole number?

A sharp reader should be uneasy here. 2^3 means “multiply three 2s together.” But our machine is supposed to accept *any* real number. So what does $2^{1/2}$ mean? Or 2^{-3} ? You can’t multiply “half a 2 together.”

The answer is that we *extend* the meaning — the same move we’ve made since Chapter 1. We decide what these should mean by insisting the machine keep behaving consistently. The rule powers must obey is $2^a \cdot 2^b = 2^{a+b}$. If we demand this keep holding for *all* exponents, the meanings are forced:

- **Zero exponent:** $2^0 = 1$ (because $2^x \cdot 2^0$ must equal 2^x).
- **Negative exponent:** $2^{-3} = \frac{1}{2^3} = \frac{1}{8}$ — a negative exponent means *divide*.
- **Fractional exponent:** $2^{1/2} = \sqrt{2}$ (because $2^{1/2} \cdot 2^{1/2} = 2^1 = 2$, so it’s the number that squares to 2).

So a fractional exponent is secretly a root, and a negative exponent is secretly a reciprocal. The upshot: 2^x makes sense for *every* real number x , and its graph is one smooth, unbroken curve.

5.3 The laws of exponents (and why every one is true)

Let's nail down the rules as a toolkit. There are seven, but here's the good news: **they're not seven separate facts. There are really just a few, and the rest are forced.**

◆ **Why it's true** — skip on a first read, but the real brain-training

Each law below comes in two parts: the **law itself** (your toolkit) and a *Why* (the proof). On a first read you can take the bold laws and skip every *Why* — you'll be able to do all the problems with just the laws. But the proofs all come down to one homely idea — *an exponent just counts how many copies you're multiplying* — so they're satisfying brain-training.

Law 1 — Product rule: $a^m \cdot a^n = a^{m+n}$. *Why:* a^m is m copies of a ; a^n is n more. Side by side, that's $m + n$ copies. Multiplying powers *adds* the exponents, because you're pooling two piles into one. This is the master rule — the other six all flow from it.

Law 2 — Quotient rule: $\frac{a^m}{a^n} = a^{m-n}$. *Why:* dividing cancels copies. Each bottom copy cancels a top copy, leaving $m - n$. The mirror of Law 1.

Law 3 — Power of a power: $(a^m)^n = a^{mn}$. *Why:* n copies of a^m , each m copies of a , is $m \times n$ copies.

Law 4 — Power of a product: $(ab)^n = a^n b^n$. *Why:* sort the n copies of ab into a pile of a 's and a pile of b 's.

The next three are *forced consequences* of Law 1, not new rules:

Law 5 — Zero exponent: $a^0 = 1$. *Why:* $a^m \cdot a^0 = a^m$, and only 1 does nothing under multiplication.

Law 6 — Negative exponent: $a^{-n} = \frac{1}{a^n}$. *Why:* $a^n \cdot a^{-n} = a^0 = 1$, so a^{-n} is the reciprocal.

Law 7 — Fractional exponent: $a^{m/n} = \sqrt[n]{a^m}$. *Why:* $(a^{1/n})^n = a^1 = a$, so $a^{1/n}$ is the n th root.

The entire toolkit is really **one idea** — **exponents count copies** — **plus its forced consequences**. If you ever blank on Law 6 or 7 in an exam, rebuild it from Law 1. The method outlasts the memory.

5.4 The shape of growth

Read the shape of 2^x from left to right. Far to the **left**, the curve hugs down close to zero but never *touches* it — 2^{-10} is tiny but still positive. The machine

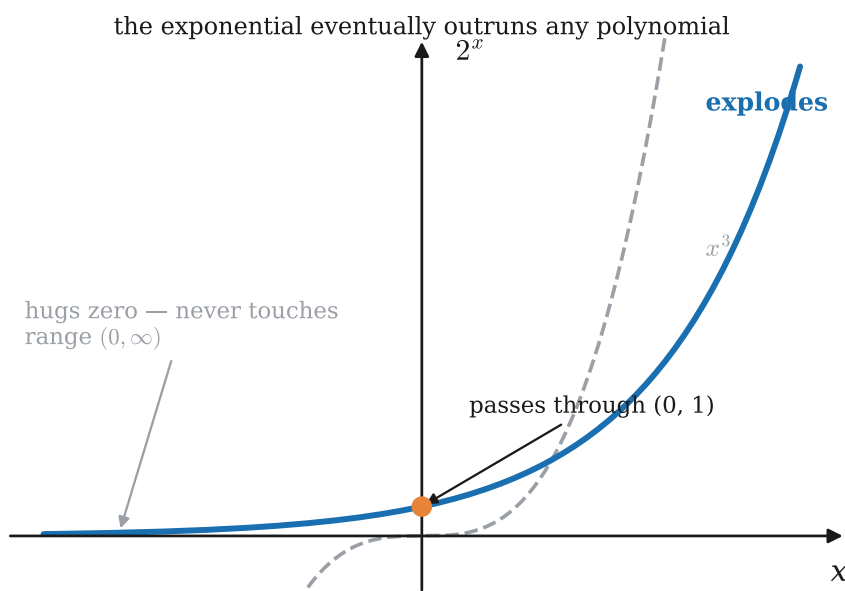


Figure 5.2. The exponential’s personality. On the left it hugs zero but never touches (range $(0, \infty)$); it passes through $(0, 1)$; on the right it explodes — and eventually outruns any polynomial, even x^3 .

can never output zero or a negative (its range is $(0, \infty)$). At $x = 0$ it passes through height 1. To the **right** it explodes — not “goes up steadily” like a line, it *accelerates*. This is **exponential growth**, and it eventually outruns *everything*. A grain of rice doubled on each square of a chessboard buries the planet.

A base *between* 0 and 1 — like $(\frac{1}{2})^x$ — does the mirror image: it *decays*. That’s **exponential decay**: radioactive atoms, a cooling drink, medicine clearing from your blood. Same machine, base flipped.

5.5 What exponentials are good for (and where they lie)

The friendly face: compound interest. Put \$100 in an account paying 10% a year. After one year, \$110. But in the second year you earn 10% on the whole \$110, gaining \$11, not \$10. Your interest earns interest. Each year multiplies your money by 1.1, so after x years you have $100 \cdot (1.1)^x$ — an exponential. This is *growth feeding on itself*, and it’s quietly where the number e comes from historically.

Now the turn: nothing compounds forever. Compound interest says \$100 becomes, after 500 years, more money than exists on Earth. Obviously that doesn’t happen. *Something always intervenes*. The same lie hides in scarier

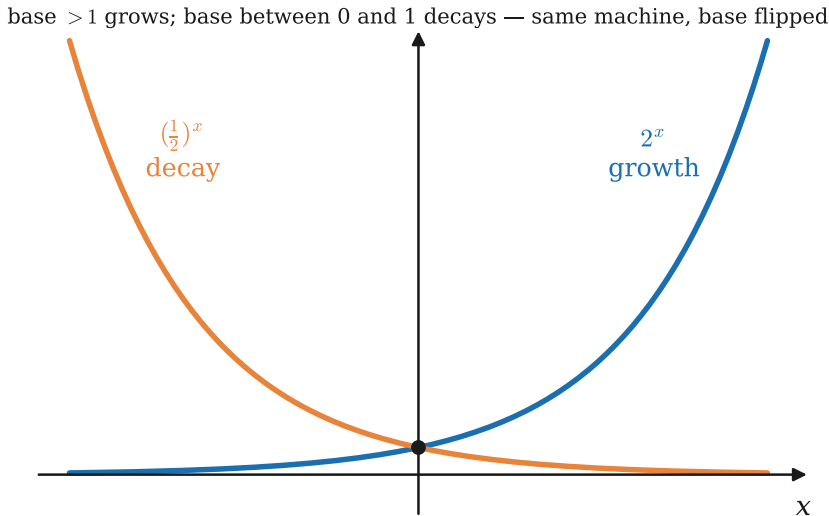


Figure 5.3. Growth versus decay, mirror images across the vertical axis. Base greater than 1 grows; base between 0 and 1 decays.

exponentials. An epidemic’s first weeks really do grow exponentially — each sick person infects two more. If that continued, within months every human would be infected several times over, which is impossible. So the growth *runs out of room*: as fewer susceptible people remain, the doubling slows, and finally stops. The curve bends over and **levels off**.

That lazy-S shape is the **logistic curve**, and it’s what exponential growth *actually* looks like in any finite world.

Exponential growth is always a model of the *early* phase. The real question is never “how fast does it grow?” but “*what is the ceiling, and when does it start to bite?*” Anyone who promises you exponential growth forever — in money, in followers, in anything — is selling you the first half of an S-curve and hiding the second.

You’ll meet this S-curve again in machine learning — and you already have every piece. The exponential decay e^{-x} heads toward zero as x grows. Build this little machine from it:

$$\sigma(x) = \frac{1}{1 + e^{-x}}.$$

When x is large and positive, e^{-x} is nearly zero, so $\sigma(x)$ is nearly 1. When x is large and negative, e^{-x} is enormous, so $\sigma(x)$ is nearly 0. At $x = 0$ we get $\frac{1}{2}$. So this machine squashes *any* input on the whole number line into the range between 0 and 1.

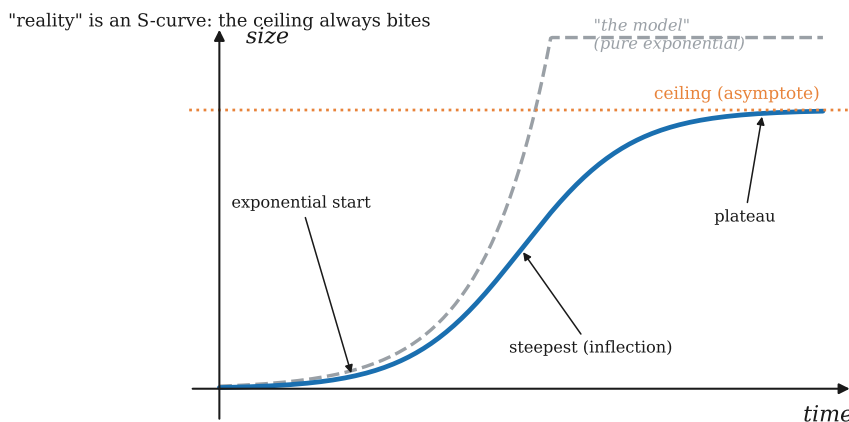


Figure 5.4. The honest correction. A pure exponential (dashed) shoots to infinity; in any finite world reality tracks the logistic *S-curve* (solid) — explosive start, steep inflection, then a plateau as the ceiling bites.

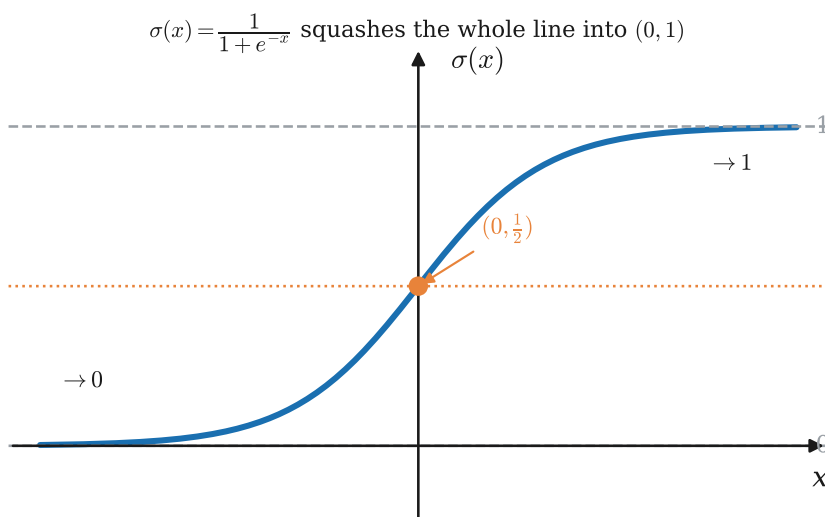


Figure 5.5. The *sigmoid* $\sigma(x) = \frac{1}{1+e^{-x}}$ — the S-curve written as a formula. It squashes the whole number line into $(0, 1)$, passing through $(0, \frac{1}{2})$. This is how an artificial neuron “decides.”

This is the **sigmoid function**, and it is exactly the S-curve. It's how an artificial neuron decides: it takes a runaway input of any size and squashes it into a bounded answer between 0 (no) and 1 (yes), with $\frac{1}{2}$ as “unsure.” The curve that tames an epidemic and the curve that lets a neural network make up its mind are the *same curve* — and you just built it from e^{-x} , a reciprocal, and a $+1$.

► **Run it yourself — drag the knobs on growth, decay, and the decision curve.** (notebook 5.1 in Colab)

Slide the base of b^x from above 1 to below 1 and watch growth flip into decay. Then watch a pure exponential race off to infinity while the S-curve bends to a ceiling. Before each slide, predict which way the curve will move — then check yourself.

5.6 Running it backwards: logarithms

Every interesting machine makes us ask the Chapter 3 question: *can we undo it?* For the exponential, the undo button is so important it gets its own name.

The doubling machine takes a number of doublings and gives the result: $3 \mapsto 8$. The *reverse* question is: “I’m looking at 8 — how many doublings produced it?” The answer is 3. That reverse machine is the **logarithm**:

$$\log_2(8) = 3, \quad \text{asking} \quad “2 \text{ raised to } \textit{what} \text{ gives } 8?”$$

A logarithm is, quite simply, *an exponent in disguise*. These two machines are perfect inverses:

$$2^3 = 8 \quad \Longleftrightarrow \quad \log_2(8) = 3.$$

This is the moment a thread from Chapter 3 pays off. We found the rule for when a machine *can* be run backwards: only if it never sends two different inputs to the same output. The squaring machine *failed* that test. But the exponential’s graph only ever *climbs* — it never repeats an output — so it passes perfectly. *That* is the deep reason the logarithm exists as a clean, single-valued undo button while the square root couldn’t.

And here’s why logarithms matter beyond being a tidy inverse: **they turn multiplication into addition**. The rule $2^a \cdot 2^b = 2^{a+b}$, read through the logarithm, becomes $\log(xy) = \log(x) + \log(y)$. For three centuries, before calculators, scientists multiplied enormous numbers by *adding* their logarithms. Logarithms are also why we measure earthquakes (Richter), loudness (decibels), and acidity (pH) the way we do — each compresses a range spanning factors of *billions* into a friendly little number.

the log is the exponential reflected across $y = x$

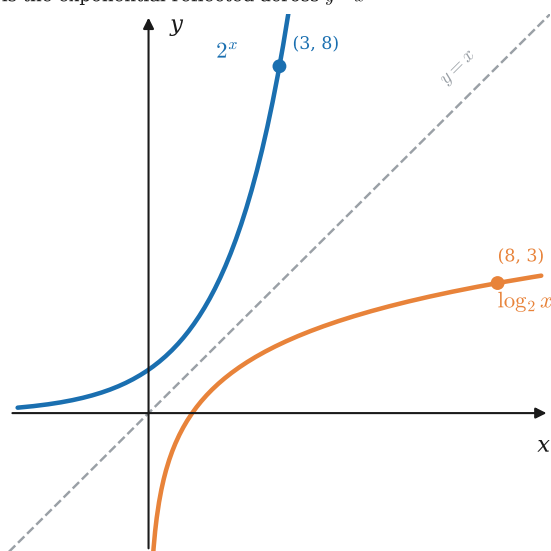


Figure 5.6. Exponential and logarithm are mirror inverses, reflected across the line $y = x$. The twin points $(3, 8)$ and $(8, 3)$ swap across the mirror.

5.7 The laws of logarithms (the exponent laws in a mirror)

Every log law is an exponent law seen through the inverse. A logarithm *is* an exponent, so everything drops down one level: *multiplication becomes addition, powers become multiplication*.

◆ Why it's true — skip on a first read, but the real brain-training

Write $x = a^p$ and $y = a^q$, so $p = \log_a x$ and $q = \log_a y$.

Law 1 — Product becomes sum: $\log_a(xy) = \log_a x + \log_a y$. *Why:* $xy = a^p \cdot a^q = a^{p+q}$, so the exponent producing xy is $p + q$.

Law 2 — Quotient becomes difference: $\log_a\left(\frac{x}{y}\right) = \log_a x - \log_a y$. *Why:* $\frac{x}{y} = a^{p-q}$.

Law 3 — Power becomes multiplier: $\log_a(x^k) = k \log_a x$. *Why:* $x^k = (a^p)^k = a^{pk}$. The most *useful* law in practice — it pulls an exponent down into an ordinary multiplier, which is how logs tame equations where the unknown is stuck up in an exponent.

Law 4 — Anchors: $\log_a 1 = 0$ and $\log_a a = 1$. *Why:* just $a^0 = 1$ and $a^1 = a$ read backwards.

The one idea to carry: a logarithm knocks every operation down a rung. Multiplication $\rightarrow$ addition. Division $\rightarrow$ subtraction. Powers $\rightarrow$ multiplication. Remember *that*, and you can regenerate the whole table.

There's one more law, the practical one, because real calculators only carry $\log_{10}$ and $\ln$ (base e). To compute $\log_2 50$ with no base-2 button, you **change the base**: $\log_a x = \frac{\log_b x}{\log_b a}$. So $\log_2 50$ is just $\frac{\ln 50}{\ln 2}$ on a calculator.

5.8 The most important question: which base?

So far we've doubled — base 2. But nothing forced that. Every base gives a perfectly good exponential. So here's a question that cracks open the rest of mathematics:

Is there a *best* base — one that makes the machine behave most simply?

To even ask that we need to say what “simply” means, and the honest answer needs calculus. But I can show you the shape of it. Every exponential curve b^x climbs, and at each point it has a *steepness*. An exponential is special: its steepness is always *proportional to its own height*. For base 2 the steepness is a bit *less* than the height; for base 3 a bit *more*. So somewhere between 2 and 3 there must be a magic base where the steepness is *exactly equal* to the height.

That magic base is an irrational number, a little past 2.7:

$$e = 2.71828\dots$$

It is called **Euler's number**, and it is one of the two or three most important numbers in all of mathematics, sharing that pedestal with π . The function e^x is the *purest* possible growth machine — the one whose growth rate, at every point, equals itself. Its inverse, $\log_e$, gets its own symbol, $\ln$ (for *natural logarithm*). I can't fully justify e until we have derivatives — but when you meet it again, you'll already know what it *is*: the number that was hiding between 2 and 3, waiting for the question “which base is best?”

5.9 Where we are now

We swapped the roles in a power and got the **exponential** machines, the mathematics of everything that grows or decays on its own terms. We met the runaway

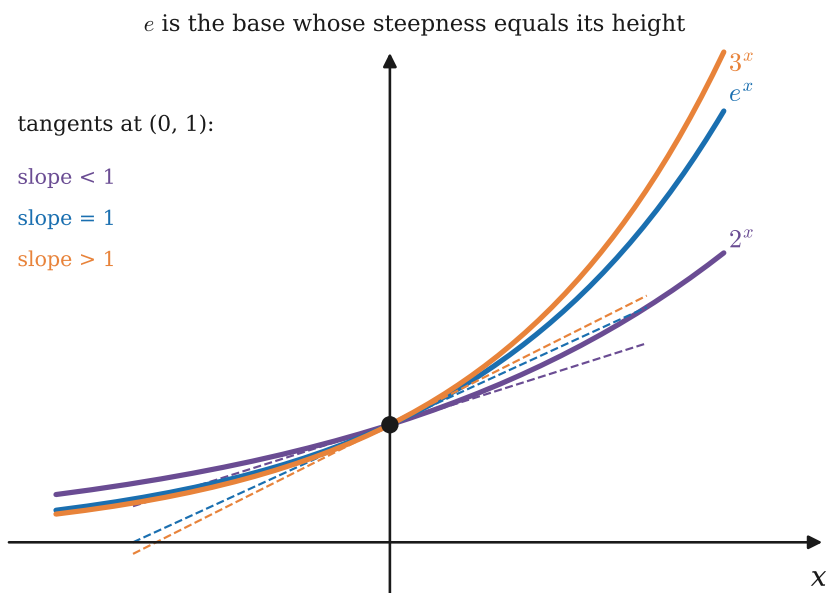


Figure 5.7. Finding e . At $(0, 1)$, the tangent to 2^x is too shallow (slope < 1) and to 3^x too steep (slope > 1). The base whose tangent has slope exactly 1 is $e \approx 2.718$ — the curve whose steepness equals its height.

personality of **exponential growth** and its mirror, **decay**. We ran the machine backwards and found the **logarithm** — an exponent in disguise, the tool that flattens multiplication into addition. And by asking “which base is best?” we found $e \approx 2.718$, the base whose growth rate exactly equals itself. We’ve now got *two* numbers (π and e) and two big promises waiting for calculus. Next we meet machines that go *around*.

Things worth thinking about

1. We found $2^{1/2} = \sqrt{2}$. Using the same logic, what should $2^{1/3}$ be? (What would you multiply together *three* times to get 2?) Connect this to the cube roots from Chapter 2.
2. The graph of 2^x gets closer and closer to zero on the left but never touches it. Why can 2^x never actually *equal* zero, no matter how negative x gets? (Can multiplying and dividing 2s ever land you exactly on nothing?)
3. A logarithm turns multiplication into addition. Before calculators, why would that single property have been worth its weight in gold to a sailor navigating by the stars?
4. (*Looking ahead.*) We claimed there’s a base e where the curve’s steepness

exactly equals its height — where the richer it gets, the faster it gets richer, in perfect proportion. Can you think of something in the real world that behaves like that?

Problems to work

Problem 1 — Forced meanings.

Using only the rule that exponents add ($2^a \cdot 2^b = 2^{a+b}$), work out what each must equal. Reason it out, don't memorize. (a) 2^0 (hint: $2^3 \cdot 2^0$ must equal 2^3); (b) 2^{-2} (hint: $2^2 \cdot 2^{-2}$ must equal 1); (c) $2^{1/2}$ (hint: $2^{1/2} \cdot 2^{1/2}$ must equal 2).

Problem 2 — Simplify with the exponent laws.

Write each as a single power of the base, naming the law(s) used. (a) $a^4 \cdot a^7$; (b) $\frac{a^9}{a^5}$; (c) $(a^3)^4$; (d) $\frac{a^2 \cdot a^6}{a^3}$; (e) $a^{-3} \cdot a^5$; (f) $(2a^2)^3$.

Problem 3 — Translate between the two languages.

Rewrite each in the other language. (a) $3^4 = 81$ as a logarithm; (b) $\log_{10}(1000) = 3$ as an exponential; (c) $\log_2\left(\frac{1}{4}\right) = -2$ as an exponential.

Problem 4 — Read the log as a question.

Answer each by asking “the base, raised to what, gives this?” — no calculator. (a) $\log_2(16)$; (b) $\log_3(9)$; (c) $\log_5(1)$; (d) $\log_2\left(\frac{1}{8}\right)$.

Problem 5 — Use the log laws.

(a) Expand $\log_a(7x)$ into a sum. (b) Expand $\log_a\left(\frac{x}{5}\right)$ into a difference. (c) Rewrite $\log_a(x^3)$ as a multiple of $\log_a x$. (d) Combine $\log_a x + \log_a y - \log_a z$ into a single logarithm. (e) (*The payoff.*) Solve $2^x = 50$. (Hint: take log of both sides, then use Law 3 to pull the x down. Leave your answer as a ratio of logs.)

★ Teach it — explain it to someone

Problem 6 — Teach it. Explain to someone the difference between x^2 and 2^x — why they're completely different machines even though both have a 2 and an x . Get them to predict which is bigger at $x = 10$ ($10^2 = 100$, but $2^{10} = 1024$), then at $x = 2$ (equal — both 4!). The fact that they *cross over* is the whole story of why exponential growth eventually wins.

Problem 7 — Compounding, and its ceiling.

\$1000 sits in an account multiplying by 1.05 each year. (a) Write the exponential function for the balance after x years. (b) Roughly how much after 1 year?

After 2? (Notice the second year's *gain* is bigger — interest earning interest.) (c) The formula says after 300 years the balance is over \$2 billion. Explain why no real deposit has ever done this, and sketch what the balance *actually* would look like. What is the “ceiling,” and what makes it bite?

Going further

★ Teach it — explain it to someone

Problem 8 — Teach it (the sigmoid decision). Explain to a family member how the curve $\sigma(x) = \frac{1}{1+e^{-x}}$ “makes a decision.” Walk them through three inputs: a big positive number (answer near 1 = “yes”), a big negative one (near 0 = “no”), and zero (exactly $\frac{1}{2}$ = “unsure”). Then drop the punchline: this exact curve is how a single artificial neuron decides, and it's *the same shape* that makes an epidemic level off. If they see one curve doing both jobs, you've taught the chapter's deepest idea.

Problem 9 — [challenge] Half-life, backwards.

A radioactive sample halves every 10 years: the amount left after t years is $A(t) = 100 \cdot (\frac{1}{2})^{t/10}$ grams. (a) How much is left after 30 years — without a calculator? (b) Now the hard direction: *after how many years is only 25 grams left?* Set $A(t) = 25$ and solve, thinking in “how many halvings?” (c) Write the exact answer for “when is only 10 grams left” using a logarithm. (This is exactly how carbon dating works.)

Problem 10 — [challenge] The log scale's superpower.

An earthquake's energy grows by a factor of about 32 for each whole step up the Richter scale. (a) How many times more energy does a magnitude-7 quake release than a magnitude-5? (b) Explain, using $\log(xy) = \log x + \log y$, why turning “multiply energy by 32” into “add 1 to the magnitude” is what makes the scale usable by humans. (c) Why would a *linear* energy scale — plain joules — be hopeless for comparing earthquakes you read about in the news?

Problem 11 — [challenge] Stacked exponents.

Which is bigger, $2^{(3^2)}$ or $(2^3)^2$? Compute both. (The lesson: exponent towers are read *top-down*, and $(a^m)^n = a^{mn}$ is *not* the same as $a^{(m^n)}$ — a trap worth feeling once with your own hands.)

CHAPTER 6

Trigonometric Functions

Every machine we've built so far has had one thing in common: feed it a bigger and bigger input, and the output runs off somewhere and never comes back. Polynomials shoot to infinity. Exponentials explode or decay to nothing. They're all machines of *escape*.

This chapter is about machines that do the opposite. You feed them a bigger and bigger input, and the output just... goes *around*. It comes back to where it started, and does it again, forever. These are the machines of **repetition** — of anything that cycles, oscillates, waves, or spins. The tides. A heartbeat. A sound wave. Alternating current. Light itself.

They're called the **trigonometric functions**, and almost everyone first meets them in a confusing way — as ratios of triangle sides, memorized from a chart. We're going to meet them the *right* way: as a point walking around a circle.

6.1 The unit circle: one walker, two shadows

Draw a circle of radius 1, centered at the origin. This is the **unit circle**. Place a walker on the rim at the rightmost point and start them walking *counterclockwise*.

Here's the whole idea of trigonometry, and it's just watching the walker's two shadows:

- The walker's **height** above the across-axis is the **sine** of the angle, $\sin \theta$.
- The walker's **across** position is the **cosine**, $\cos \theta$.

That's it. Sine and cosine are nothing more mysterious than *the height and sideways position of a point walking around a circle*. Let's read a few positions straight off the circle:

- **Start** ($\theta = 0$): rightmost point, so $\cos 0 = 1$, $\sin 0 = 0$.
- **Quarter turn up**: $\cos = 0$, $\sin = 1$.
- **Half a turn**: $\cos = -1$, $\sin = 0$.

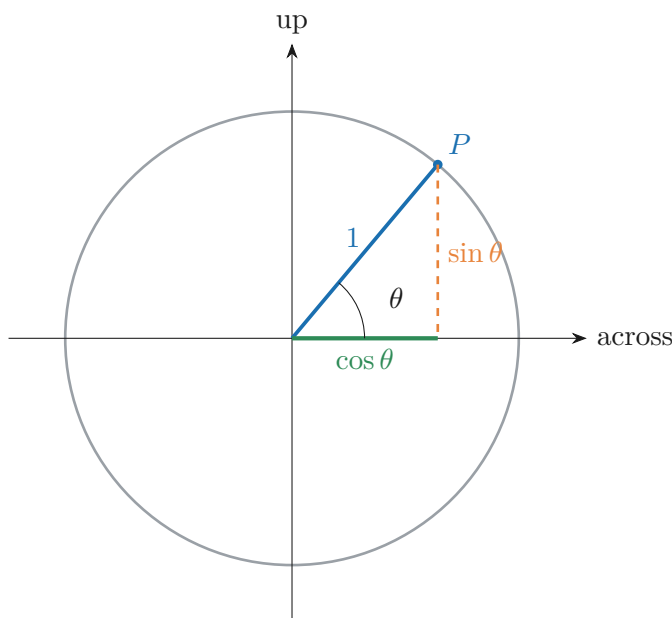


Figure 6.1. The unit-circle walker — THE foundational figure of the chapter. At angle θ , the walker’s *height* is $\sin \theta$ and its *across* position is $\cos \theta$. Everything else in trigonometry flows from this one picture.

- **Three-quarters:** $\cos = 0$, $\sin = -1$.

Notice the values never leave the range -1 to 1 — the walker is locked onto a circle of radius 1 . In Chapter 3 language: **both** $\sin$ **and** $\cos$ **have range** $[-1, 1]$. They’re *bounded* machines, the first we’ve met — utterly unlike the exponentials that ran off to infinity.

6.2 Measuring the angle: why π shows up

How do we measure θ ? You know degrees — a full turn is 360° , an ancient, arbitrary choice. Mathematicians use a more natural measure called **radians**: measure the angle by *the distance the walker has actually travelled along the rim*. Since the circle has radius 1 , its full circumference is 2π . So a full trip is 2π radians, half a trip is π , a quarter turn is $\frac{\pi}{2}$.

This is why π — the circle number from Chapter 2 — suddenly turns up *inside angle measurement*. It’s the natural unit of “going around.”

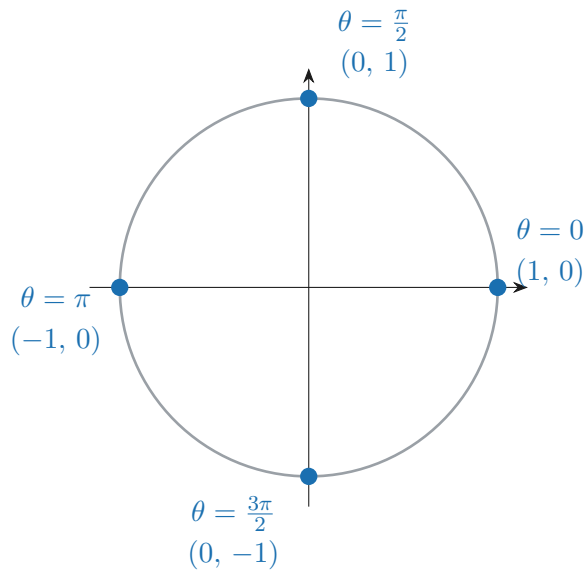


Figure 6.2. Four key positions, each labelled with its $(\cos \theta, \sin \theta)$. Read sine and cosine straight off the circle, no chart: right $\rightarrow (1, 0)$, top $\rightarrow (0, 1)$, left $\rightarrow (-1, 0)$, bottom $\rightarrow (0, -1)$.

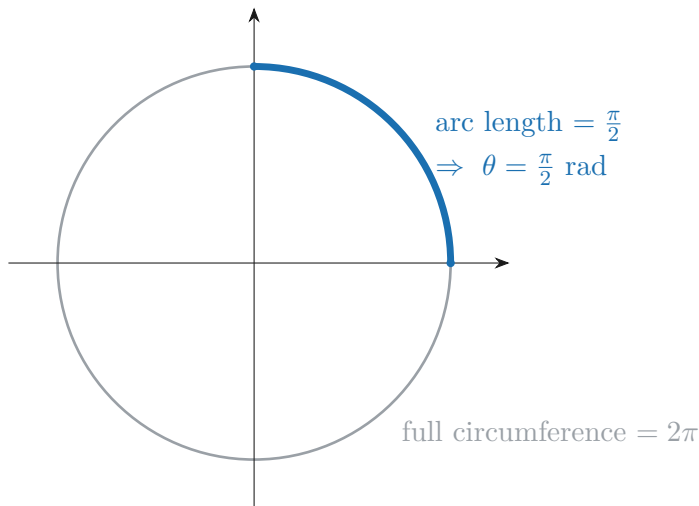


Figure 6.3. Radians as arc length. A radian measures the distance walked along the rim; a quarter turn covers arc length $\frac{\pi}{2}$, so the angle is $\frac{\pi}{2}$ radians. π is the natural unit of going around.

6.3 The special angles (built from two triangles)

To compute with sine and cosine you need their values at a handful of angles: $\frac{\pi}{6}$, $\frac{\pi}{4}$, $\frac{\pi}{3}$ (that's 30° , 45° , 60°). Most books make you memorize a table. We won't — because all six values come from just **two triangles** you can draw any time you forget them.

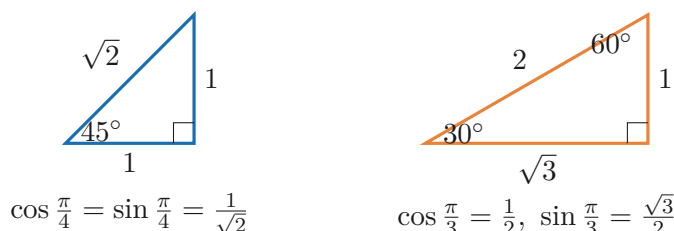


Figure 6.4. The two triangles behind every special angle. Half a unit square gives the 45° values; half an equilateral triangle (side 2) gives the 30° and 60° values. Pythagoras supplies the $\sqrt{2}$ and $\sqrt{3}$. The picture is the memory.

Triangle one — 45° , from half a square. Cut a unit square along its diagonal. Two equal short sides (both 1), two 45° angles, and a diagonal of $\sqrt{1^2 + 1^2} = \sqrt{2}$. Scale to hypotenuse 1 and read off $\cos \frac{\pi}{4} = \sin \frac{\pi}{4} = \frac{1}{\sqrt{2}} \approx 0.707$. They're equal — at 45° the walker is exactly diagonal.

Triangle two — 30° and 60° , from half an equilateral triangle. Take an equilateral triangle with all sides 2 and split it down the middle. Each half has hypotenuse 2, base 1, and height $\sqrt{2^2 - 1^2} = \sqrt{3}$. Scale to hypotenuse 1:

$$\cos \frac{\pi}{3} = \frac{1}{2}, \sin \frac{\pi}{3} = \frac{\sqrt{3}}{2}; \quad \cos \frac{\pi}{6} = \frac{\sqrt{3}}{2}, \sin \frac{\pi}{6} = \frac{1}{2}.$$

Notice $\frac{\pi}{6}$ and $\frac{\pi}{3}$ have their sine and cosine *swapped* — they're the two angles of the same triangle, seen from opposite corners.

6.4 Reading the rest off the walker: reduction identities

You'll also need to relate the sine and cosine of one angle to a related angle — things like $\sin(\pi - \theta)$. Again, no memorizing: it's just “where does the walker land if you move the angle this way?”

And the complement swap: $\frac{\pi}{2} - \theta$ reflects the walker across the 45° diagonal, which *swaps* across and up — so $\sin(\frac{\pi}{2} - \theta) = \cos \theta$. This is the deep reason *cosine* is literally “sine of the *complement*.” Don't memorize the dozen reduction formulas — hold the walker in your mind and read the answer off the circle.

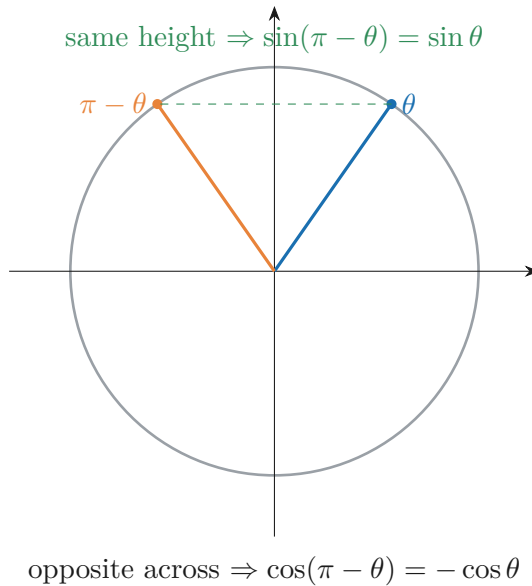


Figure 6.5. A reduction identity is a reflection. The angle $\pi - \theta$ is the mirror of θ across the vertical axis — same height, opposite across-position — so $\sin(\pi - \theta) = \sin \theta$ and $\cos(\pi - \theta) = -\cos \theta$, read straight off the circle.

6.5 The defining feature: periodicity

What happens when the walker completes a full loop and keeps going? They retrace the exact same path. The point at angle $\theta + 2\pi$ is the *same physical point* as θ . So:

$$\sin(\theta + 2\pi) = \sin \theta \quad \cos(\theta + 2\pi) = \cos \theta.$$

The machine **repeats every** 2π . This is **periodicity**, the mathematical fingerprint of everything that cycles. Plot $\sin \theta$ as the walker goes around and around, and you get the famous rolling **wave**.

That shape — the **sine wave** — is one of the most important curves in all of science. Sound is sine waves. Light is sine waves. Radio, alternating current, a plucked string, the tides — all sine waves, or sums of them.

6.6 A glimpse worth keeping

Here is the astonishing thing. Take *any* repeating shape at all — even a jagged **square wave** that jumps between two flat levels, like a light switch flicked on and off. It looks nothing like a smooth sine curve. And yet:

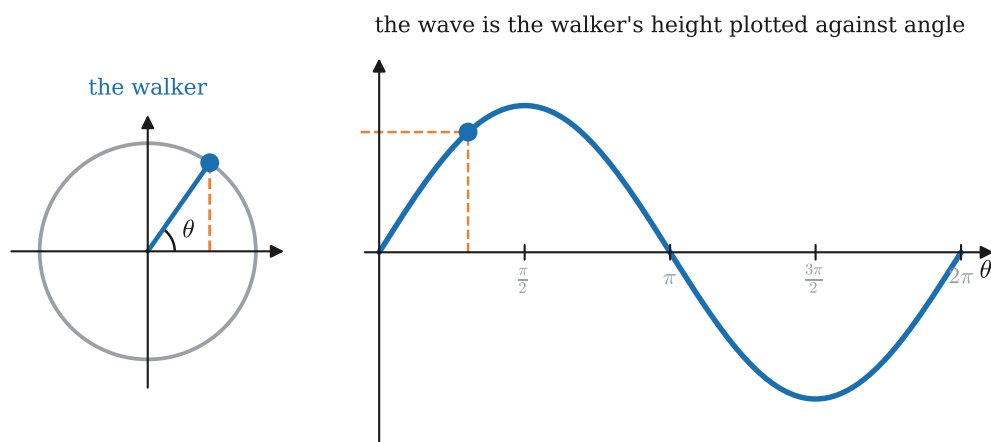


Figure 6.6. Unrolling the circle into the sine wave. The wave *is* the walker's height plotted against angle; one full lap of the circle is one full period of the wave. Cosine is the same wave with a quarter-turn head start.

Every repeating signal is secretly a **sum of pure sine waves** — a particular sine wave, plus a faster smaller one, plus a faster smaller one still, stacked up. Add the right ones and a pile of smooth curves assembles itself into a sharp-cornered square wave.

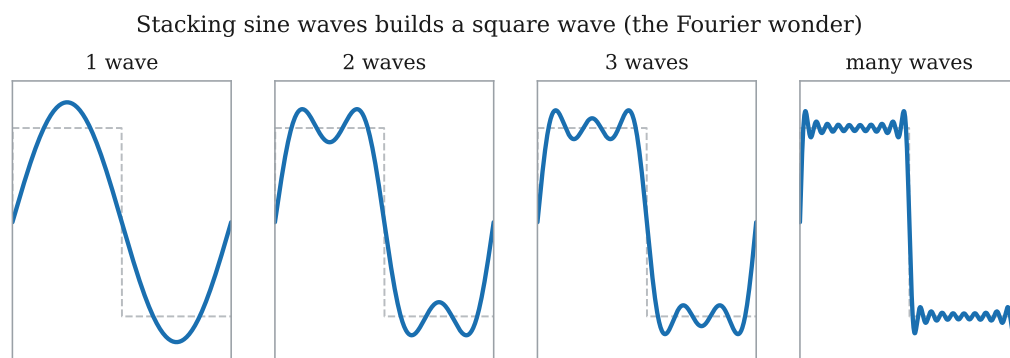


Figure 6.7. Building a square wave from sine waves — the Fourier teaser. One sine, then two, then three, then many: the smooth curves stack into sharp corners. (The recipe needs calculus; for now, just the wonder.)

The recipe for *which* sine waves is called a **Fourier expansion**, and it's the mathematics behind essentially all modern audio and images — every MP3, every JPEG. I won't teach you to compute it here, because finding the recipe needs **integration**, two books away. Hold onto the wonder: the single curve in this

chapter is the atom from which every repeating signal in the universe is built.

6.7 The third function, and the others

Sine and cosine are the two stars. The **tangent** is built from them:

$$\tan \theta = \frac{\sin \theta}{\cos \theta}.$$

Since sine is height and cosine is across-position, tangent is *height divided by across* — exactly the **slope** of the line from the center out to the walker. As the walker climbs toward straight-up, that line gets steeper and the slope shoots toward infinity.

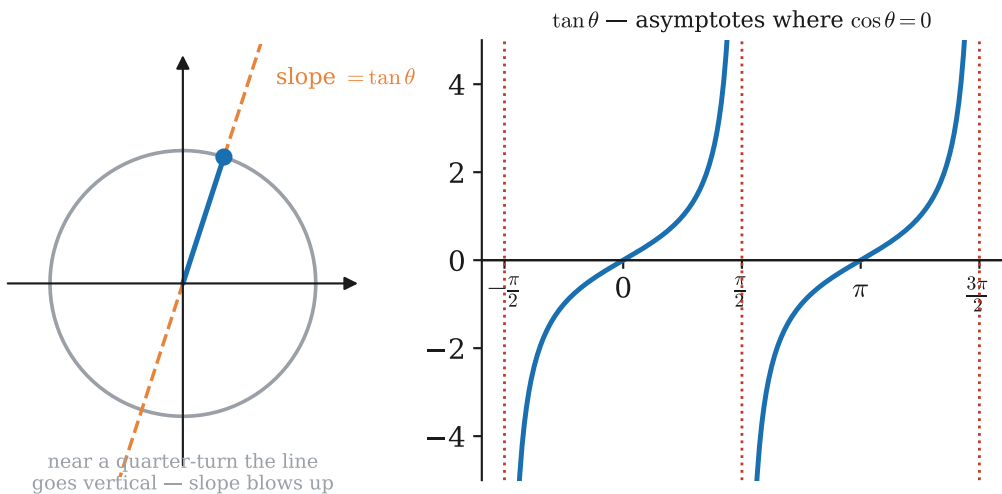


Figure 6.8. Tangent is the slope of the radius line — and it jams where $\cos \theta = 0$. As the walker nears a quarter-turn the line goes vertical and the slope blows up, giving the vertical asymptotes of $\tan \theta$.

This is the first real *domain* trouble of the chapter, and it's the Chapter 3 jam: tangent is a fraction, so it **jams wherever the bottom is zero** — wherever $\cos \theta = 0$, at every odd multiple of $\frac{\pi}{2}$. There the slope has gone vertical. The same rule governs all six trig functions:

- $\sin \theta$, $\cos \theta$ — no denominator, never jam. Domain: all angles. (This is why these two are the stars.)
- $\tan = \frac{\sin}{\cos}$ **and** $\sec = \frac{1}{\cos}$ — jam where $\cos \theta = 0$.
- $\cot = \frac{\cos}{\sin}$ **and** $\csc = \frac{1}{\sin}$ — jam where $\sin \theta = 0$.

Six functions, but really just two honest ones and four ratios — and every domain hole is nothing more than “you can’t divide by zero,” read off the circle.

6.8 The one identity to rule them all

Here is the single most important fact in all of trigonometry, and you already know *why* it’s true. Our walker is always at $(\cos \theta, \sin \theta)$, and always exactly distance 1 from the center — that’s what being on the unit circle *means*. How do you measure that distance? The **Pythagorean theorem**.

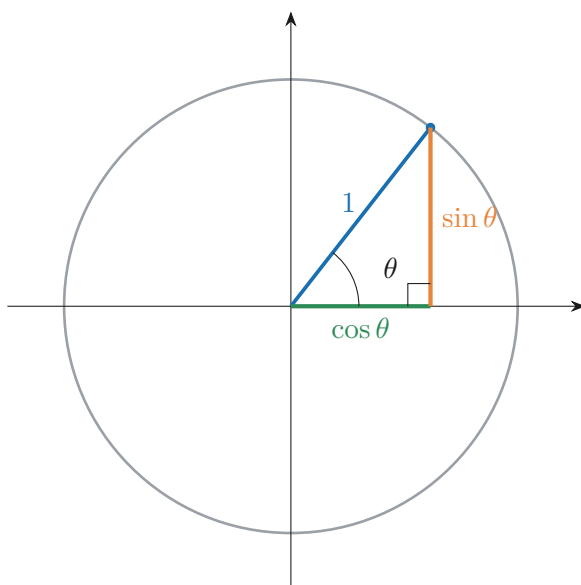


Figure 6.9. The Pythagorean identity is just Pythagoras on the radius-1 triangle. The legs are $\cos \theta$ and $\sin \theta$, the hypotenuse is the radius 1, so $\cos^2 \theta + \sin^2 \theta = 1$ for every angle, automatically.

The distance from the origin to $(\cos \theta, \sin \theta)$ is $\sqrt{\cos^2 \theta + \sin^2 \theta}$, and that distance is 1, so:

$$\cos^2 \theta + \sin^2 \theta = 1.$$

This is the **Pythagorean identity**. It’s not a new fact — it’s Pythagoras, applied to a point on a circle. Everything in trigonometry leans on it.

6.9 A toolkit of identities

Symmetry identities — walking backwards (negative angle) gives the same across-position but opposite height: $\cos(-\theta) = \cos \theta$ (cosine is *even*) and $\sin(-\theta) = -\sin \theta$ (sine is *odd*).

The addition formulas — the workhorses, giving sine and cosine of a *sum*:

$$\sin(\alpha + \beta) = \sin \alpha \cos \beta + \cos \alpha \sin \beta$$

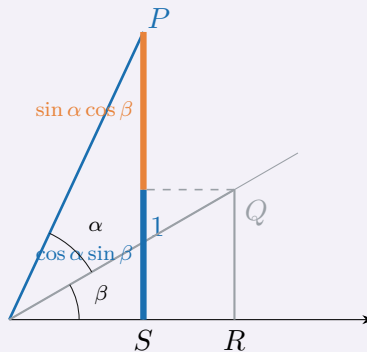
$$\cos(\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta$$

Most books simply hand these to you. We're going to *prove* them, the honest geometric way — because in Chapter 8 they'll fall out of a single magical line, but that one-line version secretly *uses* these formulas, so we need them proven independently first or the Chapter 8 argument would chase its own tail.

♦ **Why it's true** — skip on a first read, but the real brain-training

We'll prove the sine version. The whole proof is one carefully drawn picture and then *reading lengths off it*.

Stack two angles: open up β from the horizontal, then a further α on top, so the outermost ray sits at total angle $\alpha + \beta$. Mark P at distance 1 along it. Drop a perpendicular from P to the middle ray, foot Q ; then from Q , a perpendicular to the horizontal axis, foot R . What we want, $\sin(\alpha + \beta)$, is just the *height of P* .



The height of P splits into **two stacked pieces**. In right triangle OQP : $OP = 1$ and the angle at O is α , so $OQ = \cos \alpha$ and $QP = \sin \alpha$. *Piece 1* — *height of Q* : in triangle OQR , the height is $OQ \sin \beta = \cos \alpha \sin \beta$. *Piece 2* — *extra height of P above Q* : the segment QP is tilted at angle β from the

vertical, so its vertical contribution is $QP \cos \beta = \sin \alpha \cos \beta$. Add them:

$$\sin(\alpha + \beta) = \underbrace{\cos \alpha \sin \beta}_{\text{piece 1}} + \underbrace{\sin \alpha \cos \beta}_{\text{piece 2}}. \checkmark$$

The cosine version comes out the same way by reading the *horizontal* pieces, landing on $\cos(\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta$ — the minus sign appearing because the horizontal pieces work *against* each other.

The double-angle formulas — just the addition formulas with $\alpha = \beta$:

$$\sin(2\theta) = 2 \sin \theta \cos \theta \quad \cos(2\theta) = \cos^2 \theta - \sin^2 \theta.$$

You don't memorize these separately — you *derive them in one line*. One family, parent and child. The honest message: trigonometry is *not* a pile of thirty formulas. It's one picture (the walker), one identity (Pythagoras on the circle), and a couple of addition formulas.

6.10 The surprise, armed (but not yet sprung)

Think about how *strange* it is that sine and cosine have those addition formulas at all. We just proved them, so we know they're true. But the proof was fiddly — it tells you the formulas are correct without telling you *why they have that particular shape*. It looks almost like a coincidence. It is not. There's a hidden machine underneath sine and cosine, and when we meet it in Chapter 8 the addition formulas will be the *obvious*, one-line consequence of this:

$$e^{i\theta} = \cos \theta + i \sin \theta.$$

You can't read that yet — the i is the new number we haven't built, and why e (the *growth* number!) would have anything to do with circles is a genuine shock. But there it is, the most famous equation in mathematics, sitting and waiting. Hold the thread.

6.11 Where we are now

We met a completely new kind of machine — **bounded** and **periodic**, the mathematics of everything that repeats. We built **sine** and **cosine** from one honest picture: the walker on the unit circle. We saw why π governs angles, why the **sine wave** is the master shape of every oscillation, and how **tangent** is their

ratio. We found the **Pythagorean identity** — Chapter 2’s Pythagoras standing on a circle — and a small, connected toolkit, with a promise that the fearsome addition formulas are about to collapse into one beautiful line.

Things worth thinking about

1. Read off the circle, no chart: where is the walker when $\sin \theta = 1$? When $\cos \theta = -1$? Can $\sin \theta$ and $\cos \theta$ *both* equal 1 at once? (Use the Pythagorean identity — if both were 1, what would $\cos^2 + \sin^2$ come to?)
2. The sine wave repeats every 2π . If you wanted a wave that repeated every 1 unit instead, what could you do to the *input* of $\sin \theta$ to speed the walker up? (Think about $\sin(2\pi\theta)$.)
3. Tangent jams wherever $\cos \theta = 0$. Describe, in terms of the walker and the line from the center, *why* the slope becomes impossible to write down at exactly those spots. What is the line doing there?
4. (*Looking ahead.*) We claimed the *growth* number e is secretly connected to *circles*. On its face that’s absurd. What would have to be true about the new numbers of Chapter 8 for a machine that explodes and a machine that loops to turn out to be the same machine?

Problems to work

Problem 1 — Read the circle.

Without any chart, give $\sin \theta$ and $\cos \theta$ for each, saying where the walker stands. (a) $\theta = 0$; (b) $\frac{\pi}{2}$; (c) π ; (d) $\frac{3\pi}{2}$; (e) 2π (what do you notice versus $\theta = 0$?).

Problem 2 — Use the Pythagorean identity.

A walker has $\cos \theta = \frac{3}{5}$. Using $\cos^2 \theta + \sin^2 \theta = 1$, find the two possible values of $\sin \theta$. (Why *two*? One walker is above the axis, one below.)

Problem 3 — Derive a formula yourself.

Set $\alpha = \beta = \theta$ in $\sin(\alpha + \beta) = \sin \alpha \cos \beta + \cos \alpha \sin \beta$ to derive $\sin(2\theta)$. (You should get $2 \sin \theta \cos \theta$ in one line — it was hiding inside the addition formula all along.)

Problem 4 — Derive the other double-angle, three ways.

Set $\alpha = \beta = \theta$ in the cosine addition formula to get $\cos(2\theta) = \cos^2 \theta - \sin^2 \theta$. Then use the Pythagorean identity to rewrite it using only $\sin \theta$, and again using

only $\cos \theta$. (You'll find $\cos 2\theta = 1 - 2\sin^2 \theta$ and $\cos 2\theta = 2\cos^2 \theta - 1$ — three faces of one fact.)

Problem 5 — Recover the special angles from the triangles.

Without looking back, draw the two triangles and rebuild them. (a) From half a unit square, find $\sin \frac{\pi}{4}$ and $\cos \frac{\pi}{4}$. (b) From half an equilateral triangle, find $\sin \frac{\pi}{3}$, $\cos \frac{\pi}{3}$, $\sin \frac{\pi}{6}$, $\cos \frac{\pi}{6}$. If you can redraw the triangles, you own the values.

Problem 6 — Read reductions off the walker.

Use the walker picture (not memorized formulas): (a) $\sin \frac{2\pi}{3}$ (hint: $\frac{2\pi}{3} = \pi - \frac{\pi}{3}$, a reflection across the vertical); (b) $\cos \frac{2\pi}{3}$; (c) $\cos \frac{\pi}{3}$ using that it equals $\sin \frac{\pi}{6}$ (the complement swap).

Problem 7 — Derive $\tan(2\theta)$.

From $\tan(2\theta) = \frac{2\sin \theta \cos \theta}{\cos^2 \theta - \sin^2 \theta}$, divide top and bottom by $\cos^2 \theta$ and simplify, using $\tan \theta = \frac{\sin \theta}{\cos \theta}$, to show $\tan(2\theta) = \frac{2\tan \theta}{1 - \tan^2 \theta}$. (A genuine power tool for the calculus book.)

★ Teach it — explain it to someone

Problem 8 — Teach it. Explain to someone why $\sin \theta$ can never be bigger than 1 or smaller than -1 , using the walker-on-a-circle picture — *not* a formula. The cleanest explanation is purely visual: the walker is stuck on a circle of radius 1, so how high can they possibly get? If your listener sees why a *bounded* machine is fundamentally different from the exponential that ran off to infinity, you've understood what makes these functions special.

Going further

★ Teach it — explain it to someone

Problem 9 — Teach it (why cos is “co-sine”). Using only the unit circle, show a family member why $\sin(\frac{\pi}{2} - \theta) = \cos \theta$. Draw the walker at θ , then at its mirror image across the 45° diagonal, and point out that reflecting swaps “how high” with “how far across.” The punchline to land: cosine isn't a separate function to fear — it's just sine measured from the perpendicular direction. That's why the two waves look identical but shifted.

Problem 10 — State the domains.

For each function, say which angles are forbidden, using “a trig function jams where its denominator is zero” and the walker picture. (a) $\tan \theta$; (b) $\cot \theta$; (c) $\csc \theta = \frac{1}{\sin \theta}$ (same jams as which other function?); (d) why do $\sin \theta$ and $\cos \theta$ have *no* forbidden angles?

Problem 11 — [challenge] Build a wave to order.

You want a wave that oscillates between -3 and 3 (instead of -1 and 1) and repeats every π units (instead of 2π). Starting from $\sin \theta$, write the formula. (Hint: one number out front controls the height; one number *inside*, multiplying θ , controls how fast the walker goes around — and you met “inside affects the horizontal” in the transformation preview. Which inside-multiplier halves the period?)

Problem 12 — [challenge] A surprising sum.

Use the addition formulas to compute $\sin(75^\circ)$ *exactly*, without a calculator. (Hint: $75^\circ = 45^\circ + 30^\circ$, and you know the sine and cosine of both from the two triangles. Plug into $\sin(\alpha + \beta)$ and simplify to a single expression with $\sqrt{2}$ and $\sqrt{3}$.) Check that your exact answer is about 0.966 .

Problem 13 — [challenge] The identity behind a sunset.

Prove the identity $\sin^2 \theta = \frac{1 - \cos 2\theta}{2}$ starting from the double-angle formula $\cos 2\theta = 1 - 2\sin^2 \theta$ and just solving for $\sin^2 \theta$. (This innocent-looking rearrangement is one of the most-used tricks in all of calculus — it’s how you integrate $\sin^2$, which shows up everywhere from the energy in a sound wave to the blue of the sky.)

CHAPTER 7

Functions as Objects

We’ve now met every family of machine this book set out to introduce: the polynomials, the exponentials and their logarithm twins, and the trigonometric machines that go around. You could stop here and already know more about functions than most people ever learn.

But there’s one more idea, and it’s the one that turns a *collection of functions* into a *way of thinking* — the idea that quietly powers everything in the later books, from calculus to neural networks:

A function isn’t only something you *use*. It’s something you can *hold, reshape, and build with* — an object in its own right.

Up to now we’ve fed numbers into machines and watched numbers come out. In this chapter we stop feeding the machines and instead pick them up, turn them over, and *operate on the machines themselves*. The machine becomes the raw material.

7.1 Four things you can do to any function

Take any machine $f(x)$ — say the parabola $f(x) = x^2$ — and watch what four simple changes do to its graph. These four moves work *identically* on every family.

1. Shift it up or down. Add to the *output*: $f(x) + 3$ lifts the whole graph up by 3. (You’re tampering with what comes *out*.)

2. Shift it left or right. Add to the *input*: $f(x + 3)$ shifts the graph **left** by 3 — *not* right. The machine reaches its old behavior *earlier*: what f used to do at $x = 3$ now happens at $x = 0$, since $f(0 + 3) = f(3)$.

3. Stretch or squash it. $2f(x)$ doubles every height (vertical stretch); $f(2x)$ makes the machine race through its behavior twice as fast (horizontal squash).

4. Flip it. $-f(x)$ reflects across the horizontal axis (upside down); $f(-x)$ reflects across the vertical axis (mirror left-to-right).

Outside = vertical & expected • Inside = horizontal & opposite

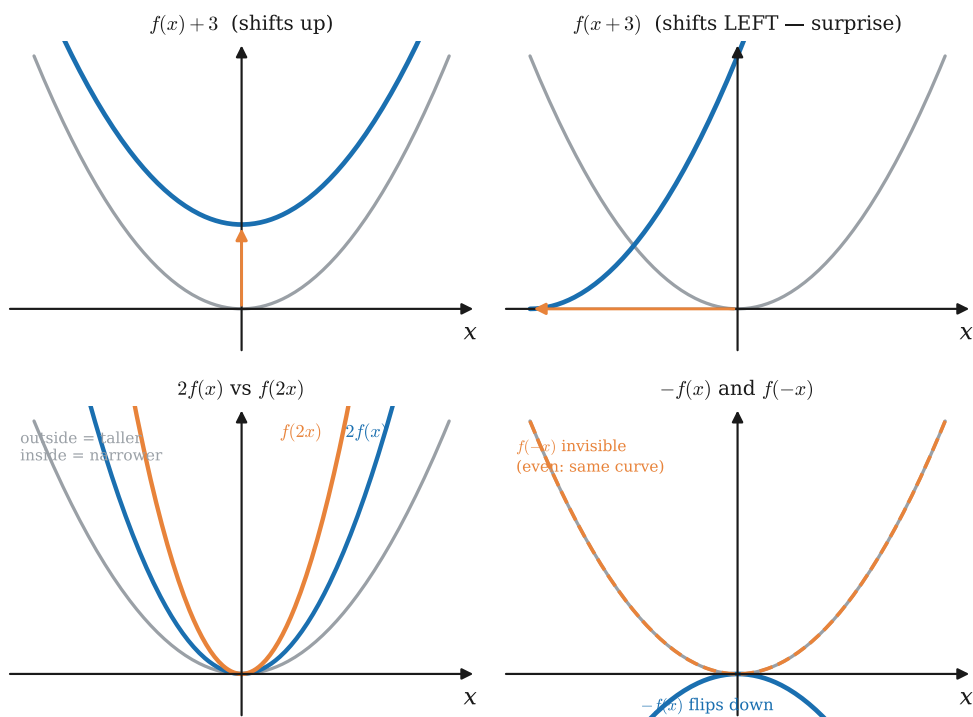


Figure 7.1. The four transformations on the base parabola (gray). Outside the function acts vertically and as expected; inside acts horizontally and in the opposite direction. (In panel D the parabola is even, so $f(-x)$ is invisible.)

Outside the function (acting on the output) changes the graph **vertically**, and does what you'd expect. **Inside the function** (acting on the input) changes the graph **horizontally**, and does the *opposite* of what you'd expect.

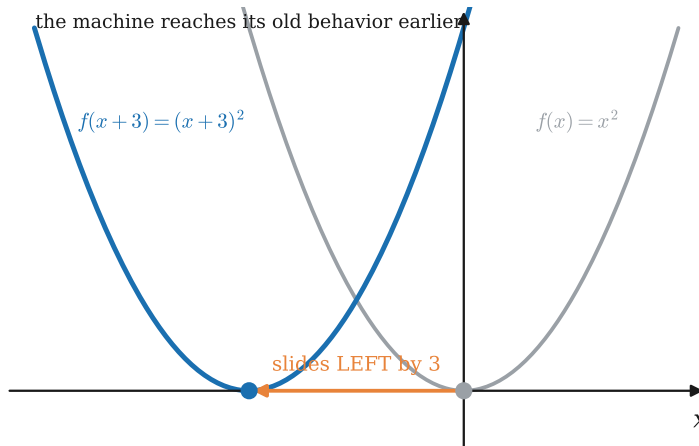


Figure 7.2. The left-shift surprise, explained. $f(x+3)$ moves the vertex from $x = 0$ to $x = -3$ — the machine reaches its old behavior earlier, so the graph slides left.

Master those two sentences and you can reshape any machine in the book. Want a sine wave twice as tall, repeating twice as fast, shifted up by one? That's $2\sin(2x) + 1$, and you can read off exactly what each piece does.

7.2 Symmetry: even and odd, properly this time

We've brushed against *even* and *odd* twice now. Now we can say what they really mean — they're statements about the *flip*. A function is **even** if flipping the input changes nothing: $f(-x) = f(x)$. Its graph is a mirror image across the vertical axis. The squaring machine is even; so is cosine. A function is **odd** if flipping the input flips the output: $f(-x) = -f(x)$. Spin it 180° about the origin and it lands on itself. The cubing machine is odd; so is sine.

Why give these names? Because symmetry is one of the most powerful labor-saving ideas in mathematics. If you know a function is even, you only have to understand *half* of it — the other half is a guaranteed mirror image, free. (And it's a quiet first taste of the symmetry that Galois built group theory from, back in Chapter 4's deep water.)

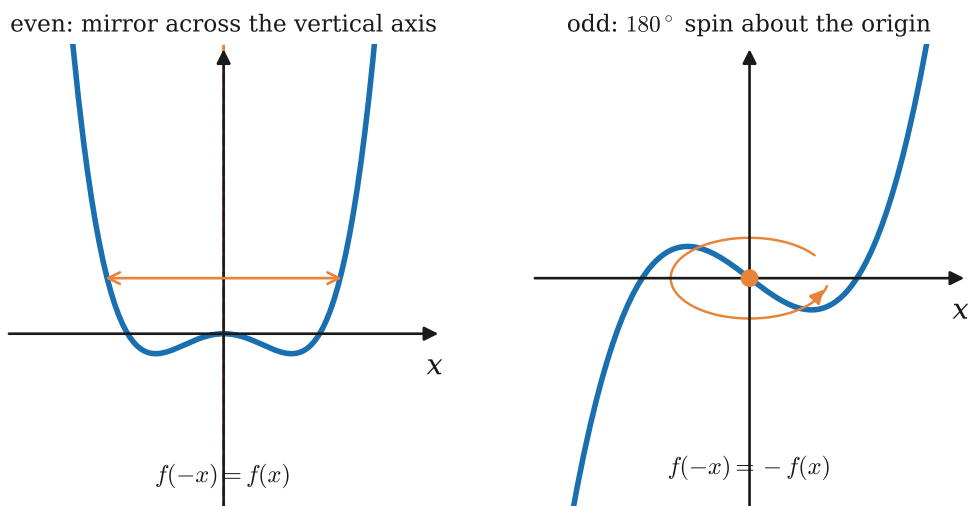


Figure 7.3. Even versus odd symmetry. Even functions mirror across the vertical axis; odd functions map onto themselves under a 180° rotation about the origin.

7.3 Building new machines from old

The transformations reshape *one* machine. The other way to make new functions is to **combine** machines. You can do arithmetic on whole functions, point by point: if $f(x) = x^2$ and $g(x) = x$, then $f + g$ is the machine $x^2 + x$. (This is how polynomials were built. Division is the one to watch — f/g jams wherever g outputs zero.)

Composition — chaining machines. This is the big one, from Chapter 3: feed the output of one machine straight into another, $f(g(x))$. I’m bringing it back here because now you can see it as an *operation that takes two machines and produces a third*. And here’s the payoff: *every transformation in the first half of this chapter is secretly a composition*. Shifting f left by 3 is composing f with the tiny “add 3” machine. Stretching is composing with a “multiply” machine.

That is precisely why composition is the heartbeat of modern mathematics. When an artificial intelligence processes something, it passes a number through an immense chain of simple machines, thousands deep. Every layer is a composition. You now hold the actual idea; the later books just stack it higher.

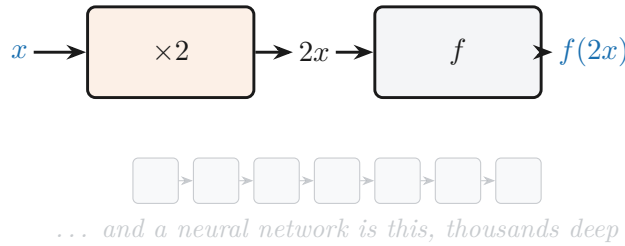


Figure 7.4. Every transformation is a composition. $f(2x)$ is “multiply the input by 2, then apply f ” — two machines in series. Stack this motif thousands deep and you have the architecture of a neural network.

7.4 Doing it again: iteration

There’s one special kind of composition so important it deserves its own name. What if you chain a machine to *itself* — feed its output straight back into its own funnel, and repeat?

$$x \rightarrow f(x) \rightarrow f(f(x)) \rightarrow f(f(f(x))) \rightarrow \dots$$

This is **iteration**, and it is one of the most powerful ideas in all of computation — *half of what computers do is iterate*. When a machine is defined in terms of itself this way, programmers call it **recursion**.

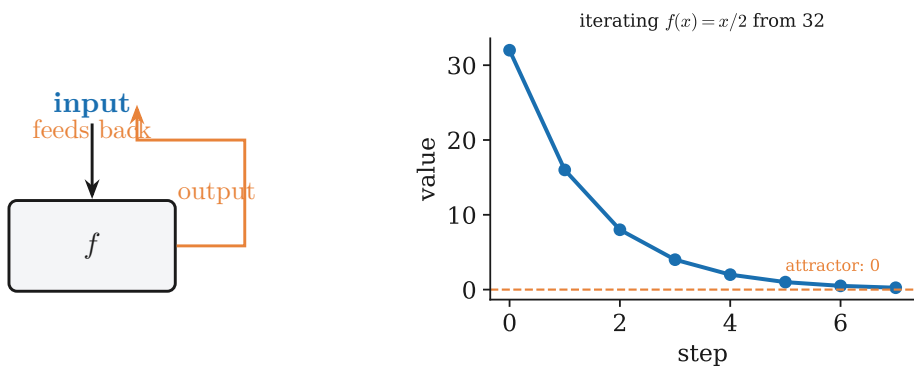


Figure 7.5. Iteration is a feedback loop: the machine’s output curves back into its own funnel. Iterating $f(x) = x/2$ from 32 marches steadily toward 0 — the *attractor*.

Let’s iterate a few machines by hand and watch the long-run behavior, because the long run is the whole point:

- **Halve it**, $f(x) = \frac{x}{2}$, from 16: 16, 8, 4, 2, 1, $\frac{1}{2}, \dots$ — marches toward 0 and never quite arrives. We call 0 an **attractor**.

- **Multiply by 2**, from 1: 1, 2, 4, 8, ... — runs off to infinity. The behavior flipped just by changing the machine.
- **Average x with $\frac{2}{x}$** , from any positive number: it homes in, startlingly fast, on $\sqrt{2}$. From $x = 1$: 1.5, then 1.4166..., then 1.41421... — the irrational number from Chapter 2, appearing as the *destination* of an iteration.

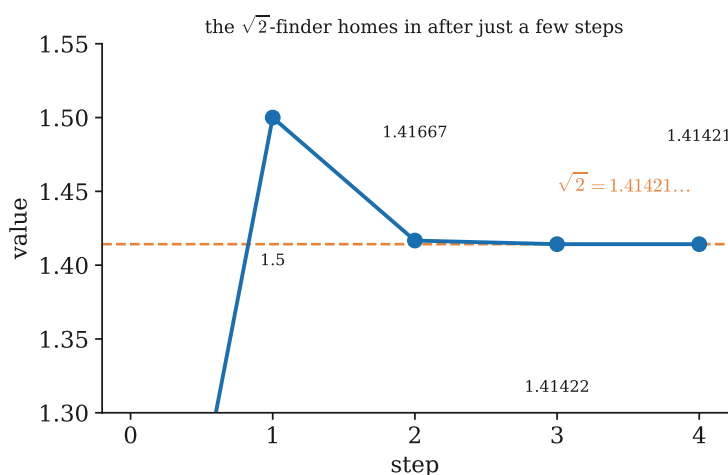


Figure 7.6. The $\sqrt{2}$ -finder converging. Iterating $f(x) = \frac{1}{2}(x + \frac{2}{x})$ lands on $\sqrt{2}$ in just a few steps — the ancestor of Newton’s method and gradient descent.

Iteration is how you *solve* things you can’t solve directly. You can’t write $\sqrt{2}$ as a fraction — but you can write a simple machine whose repeated application *converges* to it, as close as you like.

In the calculus book you’ll meet **Newton’s method**, which iterates a clever machine to hunt down the roots of *any* equation. In the machine-learning book you’ll meet **gradient descent** — the algorithm that actually *trains* every neural network — and it, too, is pure iteration: take a step downhill, look again, repeat a few million times. The training of every AI you’ve ever used is one machine, iterated. Hold that thread tightly.

► **Run it yourself** — type any number and watch it find $\sqrt{2}$. (notebook 7.1 in Colab)

Type any positive starting number and predict where the $\sqrt{2}$ -finder ends up before you run it — it always lands on $\sqrt{2}$, no matter where you start. Then swap the 2 for any other number and discover you’ve built a machine that finds square roots. This is the seed of how every AI is trained.

7.5 The proof that works like iteration: induction

Let me pause the machines to talk about something bigger. You’ve now watched iteration several times: each step refers back to the one before. There is a way of *proving* things with exactly this shape, and it’s one of the most powerful tools in all of mathematics — **proof by induction**. With it you can prove statements about *infinitely many cases at once*.

First, a word about what a *proof* even is. In every other field you gather evidence and reach a conclusion that’s *probably* right. Mathematics is the one human activity where you can be **certain**. A proof is a chain of reasons so tight that no exception can slip through — not “very likely true,” but *true, forever*. You’ve already met one kind: the proof by contradiction that cracked $\sqrt{2}$ open. Induction is the second great technique.

The domino picture. Imagine an infinite row of dominoes. Will *all* of them fall? You can’t check infinitely many. But you only need two things:

1. **The first domino falls.** (The *base case*.)
2. **Each domino, if it falls, knocks over the next.** (The *inductive step*.)

If both hold, *all* the dominoes fall — the wave runs forever.

Let’s actually do one. If you add up the first n whole numbers, is there a shortcut? The claim:

$$1 + 2 + 3 + \cdots + n = \frac{n(n+1)}{2}.$$

Base case ($n = 1$): left side is 1; right side is $\frac{1 \cdot 2}{2} = 1$. They match. The first domino falls. ✓

Inductive step: suppose it’s true for $n = k$ — assume $1 + 2 + \cdots + k = \frac{k(k+1)}{2}$. Look at the next case, $n = k + 1$, which is the same sum plus one more term:

$$\underbrace{1 + 2 + \cdots + k}_{\text{assumed}} + (k+1) = \frac{k(k+1)}{2} + (k+1) = (k+1) \left(\frac{k}{2} + 1 \right) = \frac{(k+1)(k+2)}{2}.$$

And that is *exactly* the original formula with $k + 1$ in place of n . So if it holds at k , it's forced to hold at $k + 1$. The domino at k knocks over the one at $k + 1$. ✓ Both facts established, the formula is true for *every* whole number n . We just proved infinitely many statements with six lines of algebra.

(There's a lovely legend that Gauss discovered this sum as a child, when a teacher told the class to add 1 through 100 to keep them busy, and Gauss had the answer — 5050 — almost instantly. Check: $\frac{100 \cdot 101}{2} = 5050$.)

7.6 When iteration goes wild: a doorway to chaos

Now something that genuinely unsettled mathematicians in the 1970s — and it needs nothing more than the arithmetic you already have. Take the simplest interesting feedback machine, the **logistic map** (the iterated cousin of Chapter 5's S-curve):

$$f(x) = r x (1 - x).$$

Think of x as a population (a fraction between 0 and 1). The $r x$ part is growth; the $(1 - x)$ part is the ceiling. One knob, r , controls how fast it tries to grow. We iterate: this year's population in, next year's out, repeat. Here's the unsettling part. *Just by turning the single knob r* , the long-run behavior changes in sudden, violent jumps:

- **Small r ($r = 2$):** settles to a single steady value. Calm.
- **Turn r up (around 3):** the steady value *splits in two* — now it bounces forever between a high year and a low year.
- **Turn it up more:** two splits into four, then eight — a cascade.
- **Past about $r = 3.57$:** no pattern at all. The population jumps around forever, never repeating — and yet there's no randomness anywhere. The rule is fixed and simple; you could compute every step with a pencil. But the behavior is, for all practical purposes, **unpredictable**.

This is **chaos**: a simple, fully deterministic rule producing behavior so sensitive it can't be predicted in practice. It overturned a comfortable assumption held for centuries — that simple causes make simple effects. (It's also why weather can't be forecast far ahead: tiny differences now become huge differences later. The “butterfly effect.”)

One knob turns calm into chaos — same simple rule

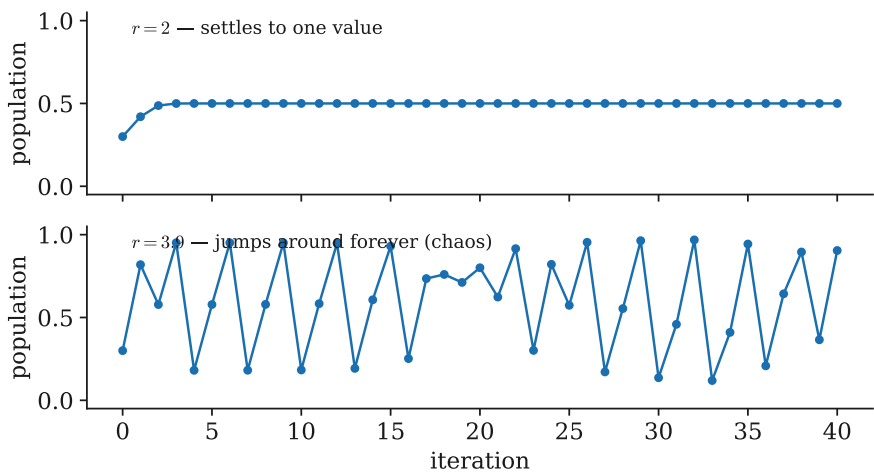


Figure 7.7. One knob turns calm into chaos. At $r = 2$ the population settles to a single value; at $r = 3.9$ the same simple rule jumps around forever with no pattern.

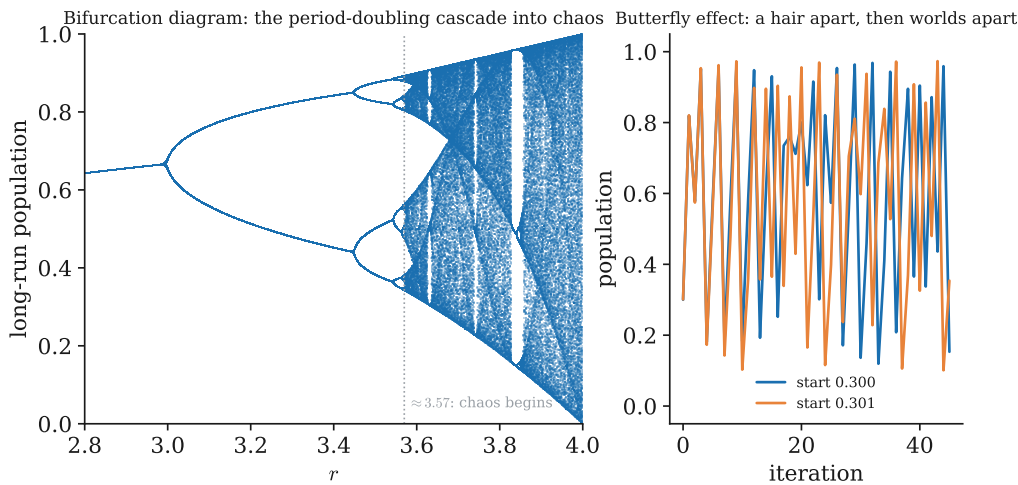


Figure 7.8. Left: the bifurcation diagram — the iconic period-doubling cascade into chaos. Right: the butterfly effect — two runs starting a hair apart (0.300 and 0.301) track closely, then explode into different futures.

► **Run it yourself — turn the knob and watch order collapse into chaos.**
(notebook 7.2 in Colab)

Drag r upward and watch the population settle, then split, then split again, then dissolve into chaos — then generate the famous bifurcation diagram that shows the whole cascade at once. Finally, start two runs a hair apart (0.300 and 0.301) and watch them explode into different futures. Do this *before* you read the next paragraph — experience it first, then read what it means.

Why I’m planting this seed now. Look at the *shape* of what happened: as you slowly turned one knob, the system held steady, held steady — then, at a sharp threshold, *abruptly reorganized* into something qualitatively new. That pattern — *a smooth change in a setting producing a sudden jump in behavior* — is called a **phase transition**, and once you have the eyes for it you see it everywhere: water that cools degree by degree and then, at one temperature, abruptly freezes; and — this is the live frontier — an artificial intelligence that is trained and trained showing no sign of a skill, and then, rather suddenly, *acquiring* it. The way large AI models seem to snap into new abilities as they grow is, many suspect, this same phenomenon. Nobody fully understands it yet. It is some of the most important unsolved territory in modern science, and it lives right here, in the doorway you’re standing in — a child iterating $rx(1-x)$ with a pencil is touching the same mathematics.

7.7 The five families, one viewpoint

Step back. We have five families of machine — **polynomials**, **exponentials**, **logarithms**, **trigonometric**, and the rational functions and combinations built from them — and a single viewpoint that unifies them: each is a *machine* (Chapter 3) with a domain and range, each can be *graphed*, *transformed*, *combined*, and sometimes *inverted*, and *composition* ties the whole structure together. That’s not five separate topics. It’s one idea — the function as a manipulable object — wearing five costumes.

This is the real reason we spent a whole book here before touching calculus. Calculus is not about numbers; it’s about *functions* — how they change, how they accumulate. You cannot study how a thing changes until you are completely comfortable holding that thing in your hands. That comfort is what this book was for. You now have it.

7.8 Where we are now — and the door ahead

We learned to treat a function as an *object*: to **shift**, **stretch**, and **flip** it, to read its **symmetry**, to **combine** machines, and to **compose** them — the master operation that scales up into the architecture of artificial intelligence. We met composition’s most potent form, **iteration**: the engine of recursion, of how we *solve* what we can’t solve directly, and the doorway to **chaos** and the **phase transition**. Along the way we met **induction**, the domino argument that proves infinitely many cases at once.

There’s just one thing left unfinished — the oldest debt in the book. Since Chapter 1 we’ve extended our numbers every time an operation broke. Then, in Chapter 4, we hit a wall and *stopped*: the quadratic formula demanded the square root of a negative, and we had no number for it. In the final chapter we open that door. We invent the number that squares to -1 , and watch — all at once — the negative discriminant resolve, every polynomial gain its full set of roots, the trig addition formulas collapse into one line, and the growth machine and the going-around machine turn out to be the same machine. Let’s finish what Chapter 1 started.

Things worth thinking about

1. $f(x + 3)$ shifts the graph *left*, which feels backwards. Check it concretely: if $f(x) = x^2$ has its lowest point at $x = 0$, where is the lowest point of $f(x + 3) = (x + 3)^2$? Solve for the input that makes the inside zero. Left or right?
2. The squaring machine is even and cubing is odd. What about $f(x) = x^2 + x$ — even, odd, or neither? (Work out $f(-x)$ and compare. Not everything has to be one or the other.)
3. We said every transformation is secretly a composition. The stretch $2f(x)$ is f composed with which simple machine — and does that machine act *before* or *after* f ? What about $f(2x)$?
4. (*Looking ahead.*) We’re about to invent a number that squares to -1 and claim it makes e^x and sine/cosine the same thing. Make a guess: what could a number that squares to a *negative* have to do with going around in a *circle*?

Problems to work

Problem 1 — Read the transformations.

Starting from $f(x) = x^2$, describe each new graph — which way it shifted,

stretched, or flipped, and by how much. (a) $f(x) - 5$; (b) $f(x - 2)$ (careful — inside!); (c) $3f(x)$; (d) $f(-x)$ (why is the answer “nothing visible?”); (e) $f(x - 1) + 4$ (two moves at once).

Problem 2 — Build a wave to order.

Starting from $\sin(x)$, write the function that is twice as tall, repeats twice as fast, and shifted up by 3. Then say which piece of your formula does each job.

Problem 3 — Even, odd, or neither.

For each, compute $f(-x)$ and decide. (a) $f(x) = x^4$; (b) $f(x) = x^3 - x$; (c) $f(x) = x^2 + 1$; (d) $f(x) = x + 1$ (show why it's neither).

Problem 4 — Iterate by hand.

Apply each machine five times from the given start, writing the sequence, then say what it does in the long run. (a) $f(x) = \frac{x}{2}$ from 32; (b) $f(x) = x + 3$ from 0; (c) the square-root finder $f(x) = \frac{1}{2}\left(x + \frac{2}{x}\right)$ from 1 (three steps; compare to $\sqrt{2} = 1.41421\dots$).

Problem 5 — Watch chaos appear.

Compute this one — it's the whole point. The machine is $f(x) = rx(1 - x)$; keep about 3 decimals. (a) Set $r = 2$, start at $x = 0.3$, iterate ~ 8 times. What value does it settle to? (b) Set $r = 3.9$, start at $x = 0.3$, iterate ~ 12 times. Does it settle, repeat, or jump around? (c) Still $r = 3.9$, start at $x = 0.301$ (a hair away) and compare to (b). Do the runs stay close or drift apart? (That drift is the butterfly effect, with your own hands.)

Problem 6 — Your first induction proof.

Prove by induction that the sum of the first n odd numbers is n^2 : $1 + 3 + 5 + \dots + (2n - 1) = n^2$. (Try a few first: $1 = 1$, $1 + 3 = 4$, $1 + 3 + 5 = 9$ — squares!) Do the base case ($n = 1$), then the inductive step: assume it holds for k , add the next odd number ($2k + 1$), and show you land on $(k + 1)^2$.

★ Teach it — explain it to someone

Problem 7 — Teach it. Explain to someone why $f(x + 3)$ shifts a graph to the *left* and not the right — the thing that confuses nearly everyone. The cleanest explanation isn't a rule; it's an example: show them that whatever f used to do at $x = 3$, the new machine now does at $x = 0$, because you fed it $0 + 3$. If you can make someone *see* why “adding inside makes it happen earlier, so the graph slides left,” you understand transformations

better than someone who just memorized “inside is backwards.”

Going further

★ Teach it — explain it to someone

Problem 8 — Teach it (the domino proof). Teach a family member how induction proves something for *all* numbers at once, using real dominoes (or books stood on end). Knock the first one (“the base case”), then explain the one rule that matters: each domino must be close enough to knock the next (“the inductive step”). Then connect it to the formula $1 + 2 + \cdots + n = \frac{n(n+1)}{2}$: checking $n = 1$ is tipping the first domino; showing “if it works for k it works for $k + 1$ ” is the spacing that guarantees the whole line falls. If they see why you *don’t* have to check every number, they’ve understood the most powerful proof technique in mathematics.

Problem 9 — [challenge] A second induction.

Prove by induction that $1 + 2 + 4 + 8 + \cdots + 2^{n-1} = 2^n - 1$ (the doubling sum from Chapter 4, now proven airtight). Base case $n = 1$: the left side is just 1 and $2^1 - 1 = 1$. ✓ For the step, assume it holds for k , add the next term 2^k to both sides, and show the right side becomes $2^{k+1} - 1$. (You’ll use $2^k + 2^k = 2^{k+1}$ — the key move.)

Problem 10 — [challenge] Find the fixed point.

A *fixed point* of a machine is an input it leaves unchanged: $f(x) = x$. (a) Show that iterating $f(x) = \frac{x}{2}$ has fixed point 0 — and that’s exactly the attractor it crawls toward. (b) The $\sqrt{2}$ -finder $f(x) = \frac{1}{2}\left(x + \frac{2}{x}\right)$ lands on $\sqrt{2}$. Confirm $\sqrt{2}$ is a fixed point by checking that $f(\sqrt{2}) = \sqrt{2}$. (c) Explain in words why an iteration can only “settle” on a value that is a fixed point. (This is the secret behind *why* these machines converge where they do — and it’s the heart of Newton’s method.)

Problem 11 — [challenge] Compose to undo.

If g is the inverse of f , then composing them gives back the input: $f(g(x)) = x$ and $g(f(x)) = x$. Take $f(x) = 3x - 6$. (a) Find g (the undo machine). (b) Verify by computing $f(g(x))$ and $g(f(x))$ and watching both collapse to x . (c) Explain why “composing a machine with its inverse gives the do-nothing machine” is just the formal version of “undo the steps in reverse order.”

Induction practice

Induction is worth drilling — it's the tool you reach for whenever something must be true *for every* n . Each of these is a short proof: nail the **base case**, assume the claim holds at k , then push to $k + 1$. (You've already done two, in Problems 6 and 9 — here are a few more, in new flavours.)

Problem 12 — Even numbers (warm-up).

Prove that $2 + 4 + 6 + \cdots + 2n = n(n + 1)$. It mirrors the $1 + 2 + \cdots + n$ proof from the chapter almost move for move — a good one to build the rhythm on.

Problem 13 — [challenge] Divisibility, a new flavour.

Prove that $n^3 - n$ is a multiple of 3 for every whole number n . There's no sum here — the *step* is the whole trick. Expand $(k + 1)^3 - (k + 1)$, peel off the $k^3 - k$ that's a multiple of 3 by assumption, and check what's left over is too. (*Hint: what's left is $3k^2 + 3k$ — and look, it wears its 3 on its sleeve.*)

Problem 14 — [challenge] An inequality.

Prove that $n < 2^n$ for every $n \geq 1$ — induction isn't only for equations. Base case: $1 < 2$. For the step, assume $k < 2^k$ and show $k + 1 < 2^{k+1}$. (*You'll want $2^k + 2^k = 2^{k+1}$, and the fact that $2^k \geq 1$.*)

Problem 15 — [challenge] Spot the flaw (the most important one).

Here is a “proof” that **all horses are the same colour**. *Base case*: a group of one horse is all the same colour — obviously. *Inductive step*: suppose any group of k horses is all one colour. Take any $k + 1$ horses in a row. The first k are one colour (by assumption); the last k are one colour (by assumption); the two groups overlap, so *everything* matches — all $k + 1$ are one colour. By induction, every horse on Earth is the same colour.

This conclusion is absurd, so the argument has a hole — **find it**. Where *exactly* does the inductive step fail? (*Decisive hint: test the very first link, from $k = 1$ to $k = 2$, with two actual horses. Do the “first k ” and “last k ” groups really overlap then?*) Getting this is worth more than any ten correct proofs: it teaches you to *check* an argument instead of trusting it because it looks like induction.

CHAPTER 8

Numbers That Shouldn't Exist

We've reached the end, and the end is also the oldest beginning. Everything this book planted comes due in this chapter.

Cast your mind all the way back to Chapter 1. There was a pattern, the deepest one in the whole book: *every time an operation broke, we invented the numbers that made it work*. Subtraction broke, so we built the negatives. Division broke, so we built the fractions. Taking roots broke, so we built the irrationals. And then, in Chapter 4, we hit a wall and did something we'd never done: we *stopped*. The quadratic formula handed us the square root of a negative number, and we had no number for it. We're going to open that door now — and when we do, every loose thread in this book pulls tight at once.

8.1 The wall, one more time

Here is the wall, in its simplest form. We want a number whose square is -1 :

$$x^2 = -1.$$

Among all the real numbers, there is no such number. A positive squared is positive. A negative squared is *also* positive ($(-1) \times (-1) = +1$). Zero squared is zero. Nothing on the entire real line squares to give a negative. The square-root machine jams, exactly as it did in Chapter 3.

For two and a half thousand years, this was the end of the conversation. “No solution.” But you know, by now, that this is never really the end. It's an invitation. Every other wall in this book fell the same way — not by working *harder* within the old numbers, but by *inventing* the new number the problem was crying out for. So let's do it one final time, and do it boldly.

8.2 The bold move

We *define* a new number — call it i — whose entire job is to be the thing that squares to -1 :

$$i^2 = -1.$$

That’s the whole definition. We’re not claiming to have *found* i hiding among the reals — it isn’t there. We’re *declaring* it into existence, the same way the ancients declared -2 into existence when subtraction demanded it. The letter i stands for “imaginary” — a slightly unfortunate name, a leftover insult from mathematicians who first distrusted it. There is nothing more imaginary about i than there is about -2 . You have never held negative two apples in your hand either. Both are inventions that earn their keep by making the mathematics work.

A number built using i — like $3 + 2i$ — is called a **complex number**. And here is the first sign we’ve done something right: with i in hand, **every quadratic now has roots**. Remember $x^2 + 2x + 5 = 0$ from Chapter 4, the one with no real roots? The quadratic formula gives

$$x = \frac{-2 \pm \sqrt{4 - 20}}{2} = \frac{-2 \pm \sqrt{-16}}{2}.$$

That $\sqrt{-16}$ used to be the wall. No longer: $\sqrt{-16} = \sqrt{16} \cdot \sqrt{-1} = 4i$. So $x = \frac{-2 \pm 4i}{2} = -1 \pm 2i$. Two roots, $-1 + 2i$ and $-1 - 2i$. The quadratic that “had no solutions” had two all along — we just lacked the numbers to write them down. **The wall is gone.**

8.3 Where do these numbers live? (The complex plane)

Here’s the question that turns i from a clever trick into something beautiful: *where does $3 + 2i$ sit?* The real numbers all live on a line. But a complex number has *two* independent parts, a real part and an imaginary part. Two numbers, two directions. So complex numbers don’t live on a line; they live on a **plane**.

This changes everything. A complex number isn’t a weird hybrid of a “real” thing and an “imaginary” thing — it’s just a *point in a plane*, as concrete as a dot on a map. And once complex numbers are points in a plane, every arithmetic operation becomes a *motion* in that plane — and that is where the real magic begins.

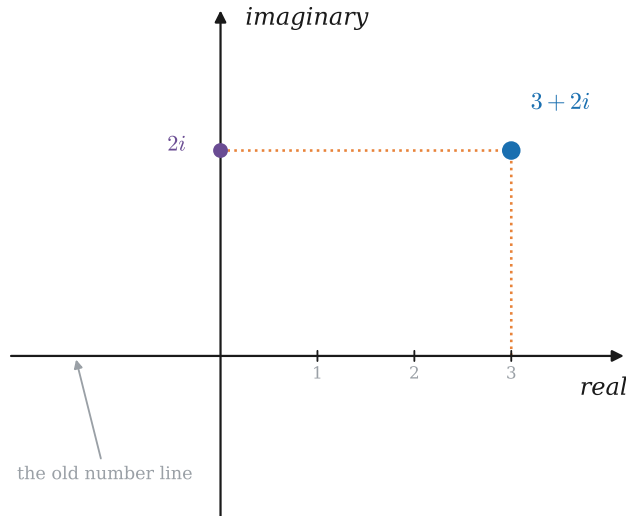


Figure 8.1. The complex plane. A complex number is just a point: $3 + 2i$ is 3 across and 2 up. The reals are the horizontal axis (the old number line); the imaginaries are the vertical axis.

8.4 Adding is sliding; multiplying is turning

Addition is the easy one: add the across-parts and up-parts separately, $(3 + 2i) + (1 + 4i) = 4 + 6i$. On the plane, that's just *sliding*, like steps on a map.

Multiplication is where the universe opens up. Multiply two complex numbers like any algebra, remembering the one rule $i^2 = -1$:

$$(a + bi)(c + di) = (ac - bd) + (ad + bc)i.$$

The *formula* is forgettable. What it *does* on the plane is not. Take any complex number and multiply it by i . Try the real number 3: $3 \times i = 3i$, which is 3 *up* the vertical axis. Multiplying by i took a point on the rightward axis and **rotated it a quarter turn**, 90° counterclockwise. Do it again: $3i \times i = 3i^2 = -3$, now pointing left. Again: $-3i$, pointing down. Again: back to 3. Four multiplications by i , four quarter-turns, all the way around and home.

Multiplying by i is a 90° rotation. And multiplying by any complex number is a *rotation combined with a scaling*. **Multiplication and turning are the same act** — the bridge we've been building toward since Chapter 2.

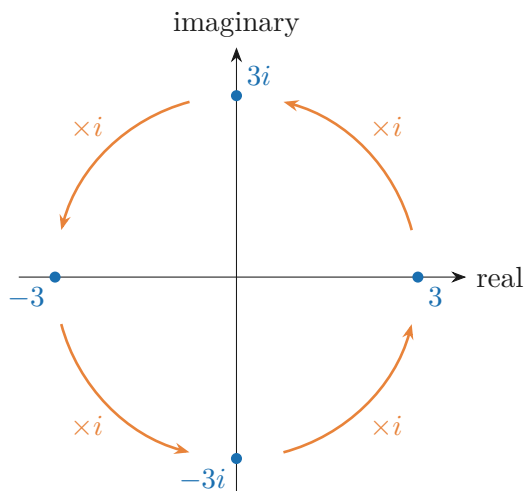


Figure 8.2. Multiplying by i is a 90° rotation. Starting from 3, each multiplication by i turns a quarter circle counterclockwise; four turns return home. The number that fixes quadratics *is* the operation of turning.

8.5 Making it rigorous: polar form, and why angles add

That claim is the heart of everything that follows, so let's *prove* it. We've been writing a complex number by its *across-and-up* address, $a + bi$. But a point in a plane has a second address: **how far it is from the origin, and which direction it points**. Call the distance r and the direction θ . This is the **polar form**.

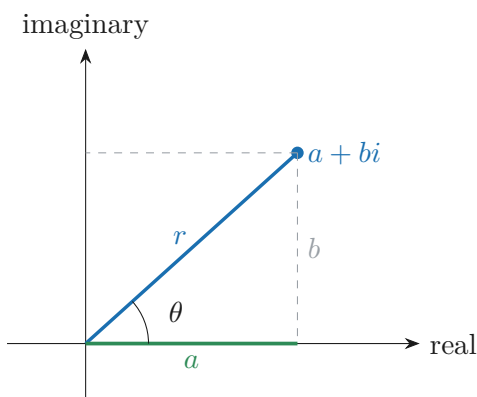


Figure 8.3. Polar form: the two addresses of one point. The across-and-up address (a, b) and the distance-and-direction address (r, θ) describe the same point, linked by $a = r \cos \theta$ and $b = r \sin \theta$.

Converting between the two addresses is pure Chapter 6. A point at distance r and angle θ has across-coordinate $r \cos \theta$ and up-coordinate $r \sin \theta$ — because cosine and sine *are* the coordinates of the unit-circle walker, scaled to radius r . So:

$$a + bi = r \cos \theta + i r \sin \theta = r(\cos \theta + i \sin \theta).$$

Every complex number is *(some distance) × (some direction)*.

♦ **Why it's true** — skip on a first read, but the real brain-training

The fact we're after is in the box below: *multiplying complex numbers multiplies their distances and adds their angles*. Take one at distance r , angle α , another at distance s , angle β . Pull the distances out front ($r \times s$) and multiply the two direction-pieces, keeping $i^2 = -1$:

$$\begin{aligned} &(\cos \alpha + i \sin \alpha)(\cos \beta + i \sin \beta) \\ &= (\cos \alpha \cos \beta - \sin \alpha \sin \beta) + i(\sin \alpha \cos \beta + \cos \alpha \sin \beta). \end{aligned}$$

Now *look* at those two grouped pieces. We have seen them before — they are exactly the right-hand sides of the **Chapter 6 addition formulas**:

$$\cos \alpha \cos \beta - \sin \alpha \sin \beta = \cos(\alpha + \beta), \quad \sin \alpha \cos \beta + \cos \alpha \sin \beta = \sin(\alpha + \beta).$$

So the whole product collapses to $rs(\cos(\alpha + \beta) + i \sin(\alpha + \beta))$ — a complex number at **distance** rs and **angle** $\alpha + \beta$. (Notice we did *not* use Euler's formula to get here — we'll need that independence in a moment.)

When you multiply two complex numbers, their distances multiply and their angles add.

8.6 The harvest: why e knows about circles

Now we spring the surprise that's been armed since Chapter 2. Hold the two facts we own:

- From Chapter 5: e^x is the **purest growth machine** — the one whose rate of change equals its own current value.
- From just now: multiplying by i means **rotating**.

Now ask the question that has no business having a nice answer: what is $e^{i\theta}$? What does the pure-growth machine do when fed an *imaginary* input?

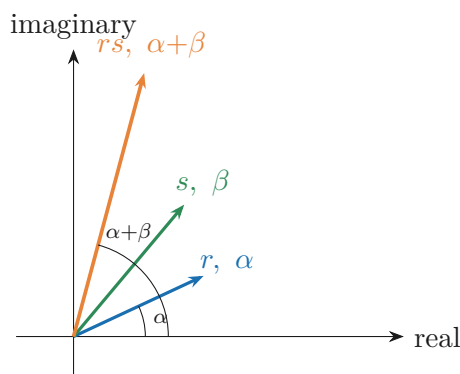


Figure 8.4. Multiplication: distances multiply, angles add. The product of arrows at angles α and β lands at angle $\alpha + \beta$ and distance rs — the rigorous form of “multiplication is rotation.”

Ordinary e^x grows in the direction it already points — straight outward. But i *turns* outward-pushing into sideways-pushing. So $e^{i\theta}$ doesn’t grow *outward* — it’s pushed *perpendicular* to where it points, at every instant. And a point always pushed exactly sideways to its own direction, never outward, does not get bigger and does not get smaller. **It goes in a circle.** (A force always perpendicular to your motion is exactly what keeps a planet in orbit.)

So $e^{i\theta}$ traces the unit circle. As θ increases, the point walks around the circle of radius 1 — and “the point on the unit circle at angle θ ” is precisely the walker from Chapter 6, whose coordinates are $(\cos \theta, \sin \theta)$. Written as a complex number, that walker is $\cos \theta + i \sin \theta$. Hence:

$$e^{i\theta} = \cos \theta + i \sin \theta.$$

There it is — **Euler’s formula**, the equation I wrote on the page back in Chapter 6 and told you that you couldn’t read yet. Now you can. It says the growth machine, fed an imaginary input, *is* the going-around machine. Exponentials and trigonometry — which looked like opposite ends of this book — are the **same machine**, seen from two directions. The bridge between them is i .

A second confirmation, pure algebra. We proved two facts from two directions: *exponents add*, $e^{i\alpha} \cdot e^{i\beta} = e^{i(\alpha+\beta)}$ (Chapter 5’s first law); and *angles add*, multiplying two unit-circle points adds their angles (just proven). These are the *same rule*. So whatever function carries angle θ to the unit-circle point at angle θ must behave like an exponential, because it obeys the exponential’s signature law. The walker and the growth machine aren’t equal by coincidence — they’re *forced* to be the same thing.

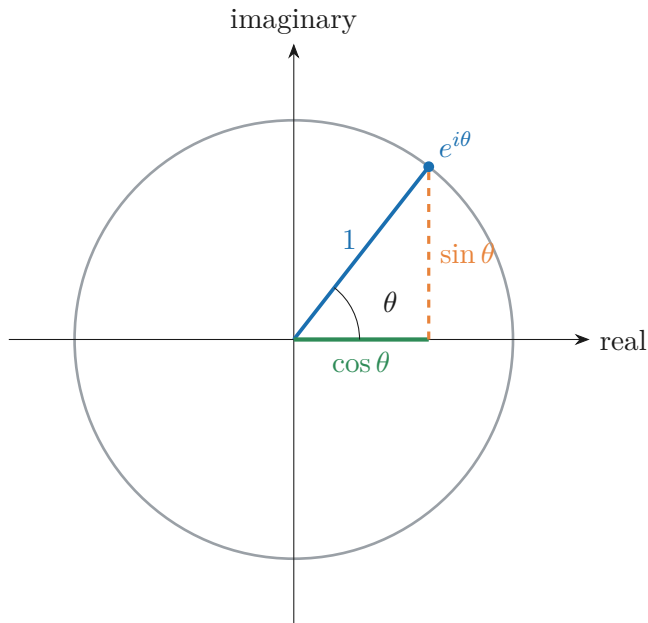


Figure 8.5. Euler's formula on the unit circle — deliberately the same picture as the Chapter 6 walker, now relabelled. $e^{i\theta}$ is the walker; its real part is $\cos \theta$ and its imaginary part is $\sin \theta$. The walker was a complex number all along.

And the most famous special case: walk *half* the circle, to $\theta = \pi$. Then $\cos \pi = -1$ and $\sin \pi = 0$, so $e^{i\pi} = -1$, usually written $e^{i\pi} + 1 = 0$ — **Euler’s identity**, tying together e , i , π , 1, and 0, the five most important numbers in mathematics, in a single line.

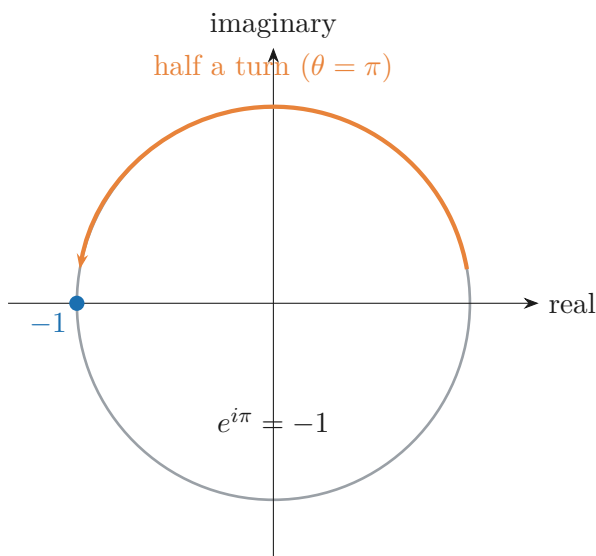


Figure 8.6. Euler’s identity. Walking half the unit circle ($\theta = \pi$) lands exactly on -1 — which is *why* $e^{i\pi} = -1$, i.e. $e^{i\pi} + 1 = 0$.

8.7 Why this also kills the addition-formula memorization

Remember the fearsome addition formulas from Chapter 6, proved the hard way with stacked triangles? They fall out of Euler’s formula in three lines. (This is *why* we proved them independently first — so the reasoning doesn’t circle.)

♦ **Why it’s true** — skip on a first read, but the real brain-training

Multiplying two “growth” expressions adds their exponents: $e^{i\alpha} \cdot e^{i\beta} = e^{i(\alpha+\beta)}$. Write each side using Euler’s formula. The right side is $\cos(\alpha + \beta) + i \sin(\alpha + \beta)$. The left side is the product of two walkers, multiplied out:

$$\begin{aligned} &(\cos \alpha + i \sin \alpha)(\cos \beta + i \sin \beta) \\ &= (\cos \alpha \cos \beta - \sin \alpha \sin \beta) + i(\sin \alpha \cos \beta + \cos \alpha \sin \beta). \end{aligned}$$

Two complex numbers are equal only when their real parts match *and* their imaginary parts match:

- Real parts: $\cos(\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta$.
 - Imaginary parts: $\sin(\alpha + \beta) = \sin \alpha \cos \beta + \cos \alpha \sin \beta$.
- Both addition formulas, at once, from one multiplication.

The thing that took a page of triangle-drawing in Chapter 6 now drops out in three lines — and you never have to memorize them again, because you can regenerate them from $e^{i\alpha} \cdot e^{i\beta} = e^{i(\alpha+\beta)}$. It was i all along.

8.8 A power tool: De Moivre, and the roots of unity made real

Euler's formula is also a *workhorse*. What is $(\cos \theta + i \sin \theta)^n$? Through Euler's formula it's $(e^{i\theta})^n = e^{i(n\theta)}$, so:

$$(\cos \theta + i \sin \theta)^n = \cos(n\theta) + i \sin(n\theta).$$

This is **De Moivre's theorem**: *raising a unit-circle point to the power n multiplies its angle by n* . A computation that would be a nightmare by repeated multiplication becomes one step — just multiply the angle.

Now let's *compute*, with our own hands, the **roots of unity** — the solutions of $x^n = 1$ that Chapter 4 only described. Take $n = 3$: the three numbers with $x^3 = 1$. Write $x = \cos \theta + i \sin \theta$ (it must sit on the unit circle). By De Moivre, $x^3 = \cos(3\theta) + i \sin(3\theta)$. We need this to equal 1 — the point at angle 0, or any full turn:

$$3\theta = 0, 2\pi, 4\pi \implies \theta = 0, \frac{2\pi}{3}, \frac{4\pi}{3}.$$

Three angles, equally spaced a third of the way around. So the three cube roots of 1 sit at 0° , 120° , 240° : the corners of a **perfect equilateral triangle**. One is the ordinary real root $x = 1$; the other two are genuinely complex.

► **Run it yourself — watch multiplication rotate, and the roots snap into a polygon.** (notebook 8.1 in Colab)

Multiply complex numbers and watch the arrows turn (distances multiply, angles add — the thing you just proved). Then set n and watch the n th roots of unity fall into the corners of a perfect n -gon. Predict the shape for $n = 6$ before you run it.

The n th roots of unity = corners of a regular n -gon on the unit circle

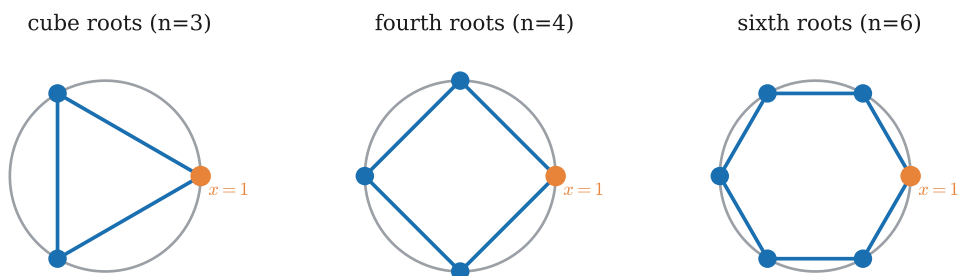


Figure 8.7. The roots of unity, computed. The n th roots of 1 are the n corners of a regular n -gon on the unit circle — Chapter 4’s promise, now derived from De Moivre. One corner is always the ordinary real root $x = 1$.

8.9 The harvest: every polynomial gets all its roots

One more debt. In Chapter 4 we noticed a degree- n polynomial has *at most* n roots, and I promised complex numbers would sharpen “at most” into “*exactly*.” Here it is, one of the great theorems of mathematics, the **Fundamental Theorem of Algebra**:

Every polynomial of degree n has exactly n roots, once you allow complex numbers (counting repeats).

Every quadratic has exactly two roots; every cubic exactly three; every degree-100 polynomial exactly one hundred. No polynomial is ever “missing” roots — the missing ones were simply complex all along, waiting off the real line for us to build the plane they live in. The real numbers were a stage too small for algebra; the complex plane is exactly the right size, and nothing bigger is ever needed.

8.10 Where we are now — and where you go next

Look at what just happened in a single chapter. We invented one number, i , with one rule, $i^2 = -1$ — and out of that one bold move, *everything we had planted came due at once*:

- the negative discriminant resolved, and **every quadratic gained its roots**;
- complex numbers turned out to be **points in a plane**, and multiplication turned out to be **rotation**;

- the growth machine and the circling machine were revealed as the **same machine**, bridged by i , in **Euler's formula**;
- the **trig addition formulas** collapsed out of it in three lines;
- and **every polynomial got its full set of roots**.

That is what this whole book was built to do: not to teach you isolated tricks, but to plant ideas early and let them grow into each other, until a single new number could tie the entire structure together. That habit — *expecting the pieces to connect, looking for the deep structure under the surface* — is the real thing you've been learning.

You started this book able to count. You end it holding the complete number system, the five great families of functions, the operations that reshape and combine them, and the most beautiful equation in mathematics — and, more importantly, knowing *why* each one had to exist. There's one machine we kept gesturing at and never built: the one that measures *how fast a function is changing*, and the one that *adds up* its accumulated effect. That machine is called **calculus**, and it's the next book. Let's keep going.

Things worth thinking about

1. We said multiplying by i rotates a point 90° . We checked it on 3. Try it on $1 + i$: compute $(1 + i) \times i$ and plot both. Did it turn a quarter circle? (Remember $i^2 = -1$.)
2. Euler's identity says $e^{i\pi} = -1$. Using the walker picture, explain *why* in one sentence: where on the unit circle does the walker stand after going angle π , and what's its complex address there?
3. The Fundamental Theorem says every degree- n polynomial has exactly n roots over the complex numbers. $x^2 + 1$ has degree 2 — so it must have 2 roots. Find them. (What squares to -1 ? There are two answers.) Where do they sit?
4. (*Looking back over the whole book.*) We extended our numbers five times: naturals $\rightarrow$ integers $\rightarrow$ rationals $\rightarrow$ reals $\rightarrow$ complex. List which operation forced each extension. Then the deep question: is there any operation left that *breaks* over the complex numbers? (The answer, for ordinary algebra, is no — the complex numbers are “complete” in a way the reals never were. That completeness is why the journey ends here.)

Problems to work

Problem 1 — Arithmetic with i .

Compute each, writing your answer as $a + bi$. The only new rule is $i^2 = -1$.
 (a) $(2 + 3i) + (4 - i)$; (b) $(2 + 3i) - (4 - i)$; (c) i^3 (hint: $i^2 \cdot i$); (d) i^4 (do powers of i cycle?); (e) $(1 + i)(1 - i)$ (watch the i^2 — the answer is purely real).

Problem 2 — Solve a quadratic with no real roots.

Use the quadratic formula to find both complex roots of $x^2 + 4x + 13 = 0$. Write them as $a + bi$ and notice they're a matched pair — same real part, opposite imaginary parts ("complex conjugates").

Problem 3 — Multiplication is rotation.

Start with 2 (the point 2 across). Multiply by i , then by i again, then again, then again — four times — writing and plotting each result. Confirm you've walked all the way around a circle and returned to 2.

Problem 4 — Use Euler's formula.

Euler's formula says $e^{i\theta} = \cos \theta + i \sin \theta$. Use it to find $e^{i\pi/2}$ (a quarter turn). What single number do you get, and where does it sit? (Check against the rotation picture — it should be straight up.)

Problem 5 — De Moivre at work.

Use $(\cos \theta + i \sin \theta)^n = \cos(n\theta) + i \sin(n\theta)$ to compute $(\cos 30^\circ + i \sin 30^\circ)^3$ *without* multiplying the long way. (Just multiply the angle by 3. What familiar point do you land on?)

★ Teach it — explain it to someone

Problem 6 — Teach it (the whole journey). This is the capstone teaching task for the entire book. Explain to someone — ideally someone who started it skeptical that "imaginary" numbers could be real — why i is no more imaginary than -2 is. Walk them through the pattern: how each kind of number (negatives, fractions, irrationals, and now i) was *invented* to fix an operation that had broken, and how i in particular turned out to secretly *be* rotation. If you can make someone see that i earned its place exactly the way negative numbers did — by making the mathematics work — then you've understood not just complex numbers, but the single idea this whole book was built around.

Going further

Problem 7 — Find the roots of unity yourself.

Following the chapter's method for the cube roots, find the four **fourth** roots of unity ($x^4 = 1$). Write $x = \cos \theta + i \sin \theta$, use De Moivre to get $x^4 = \cos(4\theta) + i \sin(4\theta)$, set it equal to 1 (angle $0, 2\pi, 4\pi, 6\pi$), and solve for the four angles. Plot them. What shape? (Check against ordinary algebra: $x^4 - 1 = (x^2 - 1)(x^2 + 1) = 0$ gives $1, -1, i, -i$. Do your angles match?)

Problem 8 — [challenge] The conjugate trick.

The *conjugate* of $a + bi$ is $a - bi$ (flip the sign of the imaginary part). (a) Show that multiplying any complex number by its conjugate, $(a + bi)(a - bi)$, always gives a *real* number — and find it (a difference of squares in disguise!). (b) Use this to divide: $\frac{1}{2+i}$. Multiply top and bottom by the conjugate $2 - i$, and write the answer as $a + bi$. (This is exactly how you divide complex numbers — by making the bottom real.)

Problem 9 — [challenge] A square root of i .

Everyone knows i is a square root of -1 . But does i *itself* have a square root — a complex number z with $z^2 = i$? Use polar thinking: i sits at distance 1, angle 90° . A square root should sit at distance 1 and *half* the angle (so that squaring — which doubles the angle — gets back to 90°). (a) What angle is that? (b) Write $z = \cos \theta + i \sin \theta$ at that angle, using your special-angle values. (c) Square it by hand to confirm you get i . (You've just taken a square root of a square root of -1 — and it was easy, because in polar form roots are just *dividing angles*.)

Problem 10 — [challenge] Why exactly n roots, seen on the circle.

The equation $x^6 = 1$ has, by the Fundamental Theorem, exactly six roots. (a) Using De Moivre, find all six angles (set 6θ equal to $0, 2\pi, 4\pi, \dots, 10\pi$). (b) Plot them — what regular shape do they make? (c) Explain in one sentence *why* there are exactly six and not more: what happens to the angle if you try 12π ? (You'll find it lands back on a root you already have — the circle only has room for n before it repeats. That's the geometric reason “degree n ” means “exactly n roots.”)